

DIV 3 CONCRETE

0301 Maintenance of Concrete

030130 Maintenance of Cast-In-Place Concrete

030130.62 CONCRETE PATCHING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Floors, 1/4" thick, small areas, regular grout	S.F.	1 Cefi	170	0.047	1.51	1.81		3.32	4.6
Epoxy grout	S.F.	1 Cefi	100	0.08	8.49	3.07		11.56	14.36
Two part polymer grout mix	S.F.	1 Cefi	150	0.053	2.15	2.05		4.2	5.69
Overhead, including chipping or sand blasting,									
Priming, and two part polymer mix, 1/4" deep	S.F.	1 Cefi	45	0.178	2.29	6.85		9.14	13.62
1/2" deep	S.F.	1 Cefi	35	0.229	4.59	8.76		13.35	19.29
3/4" deep	S.F.	1 Cefi	25	0.32	6.87	12.27		19.14	27.47
Walls, including chipping, cleaning and epoxy grout									
1/4" deep	S.F.	1 Cefi	65	0.123	7.32	4.73		12.05	15.65
1/2" deep	S.F.	1 Cefi	50	0.16	14.58	6.16		20.74	26.04
3/4" deep	S.F.	1 Cefi	40	0.2	22.1	7.7		29.8	36.54
Walls, including chipping or sand blasting,									
Priming, and two part polymer mix, 1/4" deep	S.F.	1 Cefi	80	0.1	2.29	3.84		6.13	8.73
1/2" deep	S.F.	1 Cefi	60	0.133	4.59	5.12		9.71	13.34
3/4" deep	S.F.	1 Cefi	40	0.2	6.87	7.7		14.57	20.04

030130.71 CONCRETE CRACK REPAIR

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Structural repair of concrete cracks by epoxy injection (ACI RAP-1)									
suitable for horizontal, vertical and overhead repairs									
Clean/grind concrete surface(s) free of contaminants	L.F.	1 Cefi	400	0.02		0.76		0.76	1.24
Rout crack with v-notch crack chaser, if needed	L.F.	C32	600	0.027	0.03	0.93	0.16	1.12	1.75
Blow out crack with oil-free dry compressed air (1 pass)	L.F.	C28	3000	0.003		0.11	0.01	0.12	0.18
Install surface-mounted entry ports (spacing = concrete depth)	Ea.	1 Cefi	400	0.02	1.59	0.76		2.35	2.99
Cap crack at surface with epoxy gel (per side/face)	L.F.	1 Cefi	400	0.02	0.35	0.76		1.11	1.63
Snap off ports, grind off epoxy cap residue after injection	L.F.	1 Cefi	200	0.04		1.54		1.54	2.5
Manual injection with 2-part epoxy cartridge, excludes prep									
Up to 1/32" (0.03125") wide x 4" deep	L.F.	1 Cefi	160	0.05	0.18	1.92		2.1	3.31
6" deep	L.F.	1 Cefi	120	0.067	0.27	2.56		2.83	4.44
8" deep	L.F.	1 Cefi	107	0.075	0.35	2.87		3.22	5.05
10" deep	L.F.	1 Cefi	100	0.08	0.44	3.07		3.51	5.46
12" deep	L.F.	1 Cefi	96	0.083	0.53	3.2		3.73	5.77
Up to 1/16" (0.0625") wide x 4" deep	L.F.	1 Cefi	160	0.05	0.35	1.92		2.27	3.5
6" deep	L.F.	1 Cefi	120	0.067	0.53	2.56		3.09	4.73
8" deep	L.F.	1 Cefi	107	0.075	0.71	2.87		3.58	5.44
10" deep	L.F.	1 Cefi	100	0.08	0.88	3.07		3.95	5.95
12" deep	L.F.	1 Cefi	96	0.083	1.06	3.2		4.26	6.36
Up to 3/32" (0.09375") wide x 4" deep	L.F.	1 Cefi	160	0.05	0.53	1.92		2.45	3.69
6" deep	L.F.	1 Cefi	120	0.067	0.8	2.56		3.36	5.02
8" deep	L.F.	1 Cefi	107	0.075	1.06	2.87		3.93	5.83
10" deep	L.F.	1 Cefi	100	0.08	1.33	3.07		4.4	6.43
12" deep	L.F.	1 Cefi	96	0.083	1.59	3.2		4.79	6.94
Up to 1/8" (0.125") wide x 4" deep	L.F.	1 Cefi	160	0.05	0.71	1.92		2.63	3.89
6" deep	L.F.	1 Cefi	120	0.067	1.06	2.56		3.62	5.32
8" deep	L.F.	1 Cefi	107	0.075	1.41	2.87		4.28	6.21
10" deep	L.F.	1 Cefi	100	0.08	1.77	3.07		4.84	6.92
12" deep	L.F.	1 Cefi	96	0.083	2.12	3.2		5.32	7.52
Pneumatic injection with 2-part bulk epoxy, excludes prep									
Up to 5/32" (0.15625") wide x 4" deep	L.F.	C31	240	0.033	0.25	1.27	1.39	2.91	3.86
6" deep	L.F.	C31	180	0.044	0.37	1.71	1.85	3.93	5.21
8" deep	L.F.	C31	160	0.05	0.5	1.92	2.08	4.5	5.95
10" deep	L.F.	C31	150	0.053	0.63	2.05	2.22	4.9	6.45
12" deep	L.F.	C31	144	0.056	0.75	2.13	2.31	5.19	6.82
Up to 3/16" (0.1875") wide x 4" deep	L.F.	C31	240	0.033	0.3	1.27	1.39	2.96	3.92
6" deep	L.F.	C31	180	0.044	0.45	1.71	1.85	4.01	5.29
8" deep	L.F.	C31	160	0.05	0.6	1.92	2.08	4.6	6.06
10" deep	L.F.	C31	150	0.053	0.75	2.05	2.22	5.02	6.58
12" deep	L.F.	C31	144	0.056	0.9	2.13	2.31	5.34	6.99
Up to 1/4" (0.1875") wide x 4" deep	L.F.	C31	240	0.033	0.4	1.27	1.39	3.06	4.03
6" deep	L.F.	C31	180	0.044	0.6	1.71	1.85	4.16	5.46
8" deep	L.F.	C31	160	0.05	0.81	1.92	2.08	4.81	6.28
10" deep	L.F.	C31	150	0.053	1	2.05	2.22	5.27	6.86
12" deep	L.F.	C31	144	0.056	1.2	2.13	2.31	5.64	7.33
Non-structural filling of concrete cracks by gravity-fed resin (ACI RAP-2)									
suitable for individual cracks in stable horizontal surfaces only									
Clean/grind concrete surface(s) free of contaminants	L.F.	1 Cefi	400	0.02		0.76		0.76	1.24
Rout crack with v-notch crack chaser, if needed	L.F.	C32	600	0.027	0.03	0.93	0.16	1.12	1.75
Blow out crack with oil-free dry compressed air (1 pass)	L.F.	C28	3000	0.003		0.11	0.01	0.12	0.18
Cap crack with epoxy gel at underside of elevated slabs, if needed	L.F.	1 Cefi	400	0.02	0.35	0.76		1.11	1.63
Insert backer rod into crack, if needed	L.F.	1 Cefi	400	0.02	0.02	0.76		0.78	1.27
Partially fill crack with fine dry sand, if needed	L.F.	1 Cefi	800	0.01	0.02	0.38		0.4	0.65
Apply two beads of sealant alongside crack to form reservoir, if needed	L.F.	1 Cefi	400	0.02	0.26	0.76		1.02	1.52
Manual filling with squeeze bottle of 2-part epoxy resin									
Full depth crack up to 1/16" (0.0625") wide x 4" deep	L.F.	1 Cefi	300	0.027	0.09	1.02		1.11	1.76
6" deep	L.F.	1 Cefi	240	0.033	0.14	1.27		1.41	2.22
8" deep	L.F.	1 Cefi	200	0.04	0.19	1.54		1.73	2.7
10" deep	L.F.	1 Cefi	185	0.043	0.23	1.66		1.89	2.96
12" deep	L.F.	1 Cefi	175	0.046	0.27	1.75		2.02	3.15
Full depth crack up to 1/8" (0.125") wide x 4" deep	L.F.	1 Cefi	270	0.03	0.19	1.14		1.33	2.05
6" deep	L.F.	1 Cefi	215	0.037	0.27	1.43		1.7	2.62
8" deep	L.F.	1 Cefi	180	0.044	0.36	1.71		2.07	3.17
10" deep	L.F.	1 Cefi	165	0.048	0.45	1.86		2.31	3.52
12" deep	L.F.	1 Cefi	155	0.052	0.55	1.99		2.54	3.82
Partial depth crack up to 3/16" (0.1875") wide x 1" deep	L.F.	1 Cefi	300	0.027	0.07	1.02		1.09	1.74
Up to 1/4" (0.250") wide x 1" deep	L.F.	1 Cefi	270	0.03	0.09	1.14		1.23	1.95
Up to 5/16" (0.3125") wide x 1" deep	L.F.	1 Cefi	250	0.032	0.12	1.23		1.35	2.13
Up to 3/8" (0.375") wide x 1" deep	L.F.	1 Cefi	240	0.033	0.14	1.27		1.41	2.22

CONCRETE SURFACE REPAIRS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Surface repair by Methacrylate flood coat (ACI RAP-13)									
Suitable for healing and sealing horizontal surfaces only									
Large cracks must previously have been repaired or filled									
Shotblast entire surface to remove contaminants	S.F.	A1A	4000	0.002		0.08	0.05	0.13	0.2
Blow off dust and debris with oil-free dry compressed air	S.F.	C28	16000	0.001		0.02		0.02	0.03
Flood coat surface w/Methacrylate, distribute w/broom/squeegee, no prep.	S.F.	3 Cefi	8000	0.003	0.98	0.12		1.1	1.27
Lightly broadcast even coat of dry silica sand while sealer coat is tacky	S.F.	1 Cefi	8000	0.001	0.01	0.04		0.05	0.07

Common Work Results for Concrete

Selective Demolition for Concrete

SELECTIVE DEMOLITION, CONCRETE

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Excludes saw cutting, torch cutting, loading or hauling									
Break into small pieces, reinf. less than 1% of cross-sectional area	C.Y.	B9	24	1.667		53.1	10.11	63.21	99.3
Reinforcing 1% to 2% of cross-sectional area	C.Y.	B9	16	2.5		79.65	15.19	94.84	148.39
Reinforcing more than 2% of cross-sectional area	C.Y.	B9	8	5		159.3	30.59	189.89	297.62
Remove whole pieces, up to 2 tons per piece	Ea.	E18	36	1.111		47.79	31.11	78.9	118.12
2-5 tons per piece	Ea.	E18	30	1.333		57.88	37.33	95.21	141.85
5-10 tons per piece	Ea.	E18	24	1.667		72.22	46.67	118.89	177.71
10-15 tons per piece	Ea.	E18	18	2.222		96.11	62.22	158.33	236.24
Precast unit embedded in masonry, up to 1 C.F.	Ea.	D1	16	1		35.05		35.05	58.41
1-2 C.F.	Ea.	D1	12	1.333		46.73		46.73	78.06
2-5 C.F.	Ea.	D1	10	1.6		56.29		56.29	93.99
5-10 C.F.	Ea.	D1	8	2		70.09		70.09	116.82

For hydrodemolition see Section 02 41 13.15

Basic Concrete Materials

CONCRETE ADMIXTURES AND SURFACE TREATMENTS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Abrasives, aluminum oxide, over 20 tons	Lb.				2.48			2.48	2.73
1 to 20 tons	Lb.		0		2.58			2.58	2.84
Under 1 ton	Lb.		0		2.68			2.68	2.96
Silicon carbide, black, over 20 tons	Lb.				4.24			4.24	4.66
1 to 20 tons	Lb.		0		4.41			4.41	4.85
Under 1 ton	Lb.		0		4.59			4.59	5.06
Air entraining agent, .7 to 1.5 oz. per bag, 55 gallon drum	Gal.				20.62			20.62	22.59
5 gallon pail	Gal.				28.48			28.48	31.42
Bonding agent, acrylic latex, 250 S.F./gallon, 5 gallon pail	Gal.				24.06			24.06	26.51
Epoxy resin, 80 S.F./gallon, 4 gallon case	Gal.				64.32			64.32	70.7
Calcium chloride, 50 lb. bags, T.L. lots	Ton				1202.95			1202.95	1301.15
Less than truckload lots	Bag				22.1			22.1	24.55
Carbon black, liquid, 2 to 8 lb. per bag of cement	Lb.				13.45			13.45	14.78
Concrete admixture, integral colors, dry pigment, 5 lb. bag	Ea.				27.5			27.5	30.44
10 lb. bag	Ea.				35.84			35.84	39.28
25 lb. bag	Ea.				72.67			72.67	80.03
Dustproofing compound, 250 S.F./gal., 5 gallon pail	Gal.				6.97			6.97	7.66
Epoxy based, 125 S.F./gal., 5 gallon pail	Gal.				57.94			57.94	63.83
Hardeners, metallic, 55 lb. bags, natural (grey)	Lb.				0.84			0.84	0.93
Colors	Lb.				1.45			1.45	1.6
Non-metallic, 55 lb. bags, natural grey	Lb.				0.62			0.62	0.69
Colors	Lb.				0.79			0.79	0.86
Release agent, for tilt slabs, 5 gallon pail	Gal.				17.92			17.92	19.64
For forms, 5 gallon pail	Gal.				11.64			11.64	12.77
Concrete release agent for forms, 100% biodegradable, zero VOC, 5 gal. pail	Gal.				21.11			21.11	23.08
55 gallon drum	Gal.				17.82			17.82	19.59
Sealer, hardener and dustproof, epoxy-based, 125 S.F./gal., 5 gallon unit	Gal.				57.94			57.94	63.83
3 gallon unit	Gal.				63.83			63.83	70.21
Sealer, solvent-based, 250 S.F./gal., 55 gallon drum	Gal.				21.11			21.11	23.08
5 gallon pail	Gal.				31.42			31.42	34.37
Sealer, water based, 350 S.F., 55 gallon drum	Gal.				21.11			21.11	23.08
5 gallon pail	Gal.				23.57			23.57	26.02
Set retarder, 100 S.F./gal., 1 gallon pail	Gal.				6.09			6.09	6.68
Waterproofing, integral 1 lb. per bag of cement	Lb.				3.13			3.13	3.45
Powdered metallic, 40 lbs. per 100 S.F., standard colors	Lb.				3.72			3.72	4.09
Premium colors	Lb.				5.2			5.2	5.74
For colored ready-mix concrete, add to prices in section 03 31 13.35									
Subtle shades, 5 lb. dry pigment per C.Y., add	C.Y.				27.5			27.5	30.44
Medium shades, 10 lb. dry pigment per C.Y., add	C.Y.				35.84			35.84	39.28
Deep shades, 25 lb. dry pigment per C.Y., add	C.Y.				72.67			72.67	80.03
Concrete ready mix additives, recycled coal fly ash, mixed at plant	Ton				61.87			61.87	67.76
Recycled blast furnace slag, mixed at plant	Ton				88.87			88.87	97.71

AGGREGATE

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Lightweight vermiculite or perlite, 4 C.F. bag, C.L. lots	Bag				22.1			22.1	24.06
L.C.L. lots	Bag				24.55			24.55	27.01
Sand & stone, loaded at pit, crushed bank gravel	Ton				18.9			18.9	20.62
Sand, washed, for concrete	Ton				20.62			20.62	22.59
For plaster or brick	Ton				20.62			20.62	22.59
Stone, 3/4" to 1-1/2"	Ton				17.23			17.23	18.95
Round, river stone	Ton				39.28			39.28	43.21
3/8" roofing stone & 1/2" pea stone	Ton				29.46			29.46	32.41
For trucking 10-mile round trip, add to the above	Ton	B34B	117	0.068		2.47	4.81	7.28	9.36
For trucking 30-mile round trip, add to the above	Ton	B34B	72	0.111		4.01	7.83	11.84	15.25
Sand & stone, loaded at pit, crushed bank gravel	C.Y.				26.51			26.51	28.97
Sand, washed, for concrete	C.Y.				28.97			28.97	31.92
For plaster or brick	C.Y.				28.97			28.97	31.92
Stone, 3/4" to 1-1/2"	C.Y.				32.9			32.9	36.33
Round, river stone	C.Y.				41.24			41.24	45.66
3/8" roofing stone & 1/2" pea stone	C.Y.				29.46			29.46	32.9
For trucking 10-mile round trip, add to the above	C.Y.	B34B	78	0.103		3.71	7.21	10.92	14.04
For trucking 30-mile round trip, add to the above	C.Y.	B34B	48	0.167		6.05	11.72	17.77	22.84
Onyx chips, 50 lb. bags	Cwt.				31.42			31.42	34.37
Quartz chips, 50 lb. bags	Cwt.				31.42			31.42	34.37
White marble, 3/8" to 1/2", 50 lb. bags	Cwt.				9.43			9.43	10.36
3/4", bulk	Ton				163.01			163.01	178.72

CEMENT

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Portland, Type I/II, T.L. lots, 94 lb. bags	Bag				12.37			12.37	13.6
L.T.L./L.C.L. lots	Bag				13.55			13.55	14.88
Trucked in bulk, per cwt.	Cwt.				7.66			7.66	8.45
Type III, high early strength, T.L. lots, 94 lb. bags	Bag				13.31			13.31	14.63
L.T.L. or L.C.L. lots	Bag				18.85			18.85	20.62
White, type III, high early strength, T.L. or C.L. lots, bags	Bag				26.51			26.51	29.46
L.T.L. or L.C.L. lots	Bag				37.81			37.81	41.74
White, type I, T.L. or C.L. lots, bags	Bag				27.99			27.99	30.44
L.T.L. or L.C.L. lots	Bag				29.46			29.46	31.92

WATERPROOFING AND DAMPPROOFING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Integral waterproofing, add to cost of regular concrete	C.Y.				18.81			18.81	20.62

WINTER PROTECTION

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
For heated ready mix, add	C.Y.				5.25			5.25	5.79
Temporary heat to protect concrete, 24 hours	M.S.F.	2 Clab	50	0.32	196.4	10.04		206.44	232.71
Temporary shelter for slab on grade, wood frame/polyethylene sheeting									
Build or remove, light framing for short spans	M.S.F.	2 Carp	10	1.6	299.51	64.78		364.29	436.23
Large framing for long spans	M.S.F.	2 Carp	3	5.333	402.62	215.59		618.21	797.67
Electrically heated pads, 110 volts, 15 watts/S.F., buy	S.F.				10.95			10.95	12.08
20 watts/S.F., buy	S.F.				14.58			14.58	16.06
Electrically heated pads, 15 watts/S.F., 20 uses	S.F.				0.55			0.55	0.6

Concrete Forming

Structural Cast-In-Place Concrete Forming

FORMS, BUY OR RENT

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Aluminum, smooth face, 3' x 8', buy	SFCA				14.54			14.54	16
2' x 8'	SFCA				18.87			18.87	20.78
12" x 8'	SFCA				23.58			23.58	25.83
6" x 8'	SFCA				32.01			32.01	34.81
3' x 4'	SFCA				17.69			17.69	19.48
2' x 4'	SFCA				21.51			21.51	23.58
12" x 4'	SFCA				27.51			27.51	30.32
6" x 4'	SFCA				38.18			38.18	41.55
Textured brick face, 3' x 8', buy	SFCA				17.57			17.57	19.37
2' x 8'	SFCA				24.71			24.71	27.51
12" x 8'	SFCA				35.37			35.37	38.74
6" x 8'	SFCA				55.03			55.03	60.64
3' x 4'	SFCA				21.84			21.84	24.14
2' x 4'	SFCA				29.76			29.76	33.13
12" x 4'	SFCA				42.67			42.67	46.6
6" x 4'	SFCA				66.26			66.26	73
Average cost incl. accessories but not incl. ties, buy	SFCA				25.27			25.27	28.08
Rent per month	SFCA				1.27			1.27	1.4
Metal framed plywood 2' x 8', buy	SFCA				13.14			13.14	14.49
2' x 3'	SFCA				14.49			14.49	15.89
1' x 3'	SFCA				19.71			19.71	21.73
Average cost incl. accessories but not incl. ties, buy	SFCA				17.74			17.74	19.54
Rent per month	SFCA				0.89			0.89	0.98
Plywood modular prefabricated 2' x 8', buy	SFCA				7.19			7.19	7.86
Average cost incl. accessories but not incl. ties, buy	SFCA				9.66			9.66	10.61
Rent per month	SFCA				0.97			0.97	1.07
Wood, 5/8" exterior plyform, buy	S.F.				1.72			1.72	1.89
3/4" exterior plyform	S.F.				1.39			1.39	1.54
Lumber, framing, average	M.B.F.				741.18			741.18	814.18

FORMS IN PLACE, BEAMS AND GIRDERS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Exterior spandrel, job-built plywood, 12" wide, 1 use	SFCA	C2	225	0.213	3.43	7.41		10.84	16.12
2 use	SFCA	C2	275	0.175	1.8	6.02		7.82	12.06
3 use	SFCA	C2	295	0.163	1.37	5.66		7.03	10.92
4 use	SFCA	C2	310	0.155	1.11	5.35		6.46	10.17
18" wide, 1 use	SFCA	C2	250	0.192	3.03	6.64		9.67	14.45
2 use	SFCA	C2	275	0.175	1.66	6.02		7.68	11.91
3 use	SFCA	C2	305	0.157	1.21	5.45		6.66	10.45
4 use	SFCA	C2	315	0.152	0.99	5.3		6.29	9.89
24" wide, 1 use	SFCA	C2	265	0.181	2.76	6.28		9.04	13.47
2 use	SFCA	C2	290	0.166	1.56	5.71		7.27	11.28
3 use	SFCA	C2	315	0.152	1.1	5.3		6.4	10.01
4 use	SFCA	C2	325	0.148	0.9	5.11		6.01	9.53
Interior beam, job-built plywood, 12" wide, 1 use	SFCA	C2	300	0.16	4.17	5.56		9.73	13.85
2 use	SFCA	C2	340	0.141	2.01	4.89		6.9	10.39
3 use	SFCA	C2	364	0.132	1.66	4.57		6.23	9.44
4 use	SFCA	C2	377	0.127	1.36	4.4		5.76	8.85
24" wide, 1 use	SFCA	C2	320	0.15	2.82	5.2		8.02	11.75
2 use	SFCA	C2	365	0.132	1.59	4.56		6.15	9.36
3 use	SFCA	C2	385	0.125	1.12	4.32		5.44	8.44
4 use	SFCA	C2	395	0.122	0.91	4.21		5.12	8
Encasing steel beam, hung, job-built plywood, 1 use	SFCA	C2	325	0.148	3.54	5.11		8.65	12.44
2 use	SFCA	C2	390	0.123	1.94	4.26		6.2	9.24
3 use	SFCA	C2	415	0.116	1.41	4		5.41	8.25
4 use	SFCA	C2	430	0.112	1.15	3.87		5.02	7.7
Bottoms only, to 30" wide, job-built plywood, 1 use	SFCA	C2	230	0.209	4.65	7.2		11.85	17.16
2 use	SFCA	C2	265	0.181	2.61	6.28		8.89	13.3
3 use	SFCA	C2	280	0.171	1.86	5.92		7.78	11.92
4 use	SFCA	C2	290	0.166	1.52	5.71		7.23	11.23
Sides only, vertical, 36" high, job-built plywood, 1 use	SFCA	C2	335	0.143	5.78	4.96		10.74	14.68
2 use	SFCA	C2	405	0.119	3.19	4.11		7.3	10.34
3 use	SFCA	C2	430	0.112	2.31	3.87		6.18	8.98
4 use	SFCA	C2	445	0.108	1.89	3.74		5.63	8.3
Sloped sides, 36" high, 1 use	SFCA	C2	305	0.157	5.57	5.45		11.02	15.23
2 use	SFCA	C2	370	0.13	3.1	4.5		7.6	10.92
3 use	SFCA	C2	405	0.119	2.22	4.11		6.33	9.29
4 use	SFCA	C2	425	0.113	1.81	3.91		5.72	8.52
Upstanding beams, 36" high, 1 use	SFCA	C2	225	0.213	7.13	7.41		14.54	20.21
2 use	SFCA	C2	255	0.188	3.98	6.53		10.51	15.23
3 use	SFCA	C2	275	0.175	2.89	6.02		8.91	13.25
4 use	SFCA	C2	280	0.171	2.34	5.92		8.26	12.45

FORMS IN PLACE, COLUMNS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Round fiberglass, 4 use per mo., rent, 12" diameter	L.F.	C1	160	0.2	9.15	6.89		16.04	21.58
16" diameter	L.F.	C1	150	0.213	10.89	7.36		18.25	24.27
18" diameter	L.F.	C1	140	0.229	12.18	7.87		20.05	26.54
24" diameter	L.F.	C1	135	0.237	15.22	8.13		23.35	30.31
28" diameter	L.F.	C1	130	0.246	17.01	8.49		25.5	32.8
30" diameter	L.F.	C1	125	0.256	17.74	8.8		26.54	34.14
36" diameter	L.F.	C1	120	0.267	23.58	9.16		32.74	41.11
Round fiber tube, recycled paper, 1 use, 8" diameter	L.F.	C1	155	0.206	3.23	7.1		10.33	15.39
10" diameter	L.F.	C1	155	0.206	4.05	7.1		11.15	16.29
12" diameter	L.F.	C1	150	0.213	4.55	7.36		11.91	17.26
14" diameter	L.F.	C1	145	0.221	4.82	7.61		12.43	17.96
16" diameter	L.F.	C1	140	0.229	5.95	7.87		13.82	19.69
18" diameter	L.F.	C1	140	0.229	6.96	7.87		14.83	20.81
20" diameter	L.F.	C1	135	0.237	8.7	8.13		16.83	23.13
24" diameter	L.F.	C1	130	0.246	10.84	8.49		19.33	26.06

30" diameter	L.F.	C1	125	0.256	16.28	8.8	25.08	32.57	
36" diameter	L.F.	C1	115	0.278	22.01	9.57	31.58	40.09	
42" diameter	L.F.	C1	100	0.32	48.85	11.01	59.86	72.27	
48" diameter	L.F.	C1	85	0.376	61.2	12.97	74.17	88.99	
For seamless type, add					15.00%				
Round, steel, 4 use per mo., rent, regular duty, 14" diameter	L.F.	C1	145	0.221	20.33	7.61	27.94	35.01	
16" diameter	L.F.	C1	125	0.256	20.72	8.8	29.52	37.68	
Heavy duty, 20" diameter	L.F.	C1	105	0.305	22.46	10.5	32.96	42.2	
24" diameter	L.F.	C1	85	0.376	24.71	12.97	37.68	48.56	
30" diameter	L.F.	C1	70	0.457	28.08	15.74	43.82	57.12	
36" diameter	L.F.	C1	60	0.533	30.32	18.37	48.69	63.49	
48" diameter	L.F.	C1	50	0.64	44.92	22.12	67.04	85.94	
60" diameter	L.F.	C1	45	0.711	55.59	24.7	80.29	101.85	
For second and succeeding months, deduct					50.00%				
Job-built plywood, 8" x 8" columns, 1 use	SFCA	C1	165	0.194	2.99	6.69	9.68	14.4	
2 use	SFCA	C1	195	0.164	1.71	5.66	7.37	11.31	
3 use	SFCA	C1	210	0.152	1.19	5.25	6.44	10.06	
4 use	SFCA	C1	215	0.149	0.99	5.12	6.11	9.61	
12" x 12" columns, 1 use	SFCA	C1	180	0.178	2.9	6.12	9.02	13.38	
2 use	SFCA	C1	210	0.152	1.59	5.25	6.84	10.5	
3 use	SFCA	C1	220	0.145	1.16	5	6.16	9.61	
4 use	SFCA	C1	225	0.142	0.94	4.89	5.83	9.21	
16" x 16" columns, 1 use	SFCA	C1	185	0.173	2.91	5.97	8.88	13.13	
2 use	SFCA	C1	215	0.149	1.55	5.12	6.67	10.25	
3 use	SFCA	C1	230	0.139	1.17	4.78	5.95	9.26	
4 use	SFCA	C1	235	0.136	0.95	4.68	5.63	8.86	
24" x 24" columns, 1 use	SFCA	C1	190	0.168	3.3	5.81	9.11	13.3	
2 use	SFCA	C1	216	0.148	1.81	5.09	6.9	10.49	
3 use	SFCA	C1	230	0.139	1.31	4.78	6.09	9.42	
4 use	SFCA	C1	238	0.134	1.07	4.62	5.69	8.9	
36" x 36" columns, 1 use	SFCA	C1	200	0.16	2.43	5.51	7.94	11.83	
2 use	SFCA	C1	230	0.139	1.37	4.78	6.15	9.49	
3 use	SFCA	C1	245	0.131	0.97	4.5	5.47	8.58	
4 use	SFCA	C1	250	0.128	0.79	4.4	5.19	8.22	
Steel framed plywood, based on 50 uses of purchased forms, and 4 uses of bracing lumber									
8" x 8" column	SFCA	C1	340	0.094	2.45	3.24	5.69	8.08	
10" x 10"	SFCA	C1	350	0.091	2.12	3.15	5.27	7.59	
12" x 12"	SFCA	C1	370	0.086	1.81	2.97	4.78	6.95	
16" x 16"	SFCA	C1	400	0.08	1.4	2.75	4.15	6.14	
20" x 20"	SFCA	C1	420	0.076	1.25	2.62	3.87	5.74	
24" x 24"	SFCA	C1	440	0.073	0.9	2.5	3.4	5.16	
30" x 30"	SFCA	C1	440	0.073	1.15	2.5	3.65	5.43	
36" x 36"	SFCA	C1	460	0.07	1	2.4	3.4	5.09	
FORMS IN PLACE, CULVERT									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
5' to 8' square or rectangular, 1 use	SFCA	C1	170	0.188	4.46	6.48		10.94	15.71
2 use	SFCA	C1	180	0.178	2.68	6.12		8.8	13.14
3 use	SFCA	C1	190	0.168	2.1	5.81		7.91	11.98
4 use	SFCA	C1	200	0.16	1.81	5.51		7.32	11.15
FORMS IN PLACE, ELEVATED SLABS									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Flat plate, job-built plywood, to 15' high, 1 use	S.F.	C2	470	0.102	4.35	3.54		7.89	10.69
2 use	S.F.	C2	520	0.092	2.39	3.2		5.59	7.98
3 use	S.F.	C2	545	0.088	1.74	3.05		4.79	6.99
4 use	S.F.	C2	560	0.086	1.41	2.96		4.37	6.5
15' to 20' high ceilings, 4 use	S.F.	C2	495	0.097	1.41	3.35		4.76	7.16
21' to 35' high ceilings, 4 use	S.F.	C2	450	0.107	1.77	3.69		5.46	8.11
Flat slab, drop panels, job-built plywood, to 15' high, 1 use	S.F.	C2	449	0.107	5	3.7		8.7	11.66
2 use	S.F.	C2	509	0.094	2.75	3.26		6.01	8.47
3 use	S.F.	C2	532	0.09	2	3.13		5.13	7.4
4 use	S.F.	C2	544	0.088	1.63	3.06		4.69	6.88
15' to 20' high ceilings, 4 use	S.F.	C2	480	0.1	2.86	3.47		6.33	8.92
20' to 35' high ceilings, 4 use	S.F.	C2	435	0.11	3.21	3.82		7.03	9.92
Floor slab hung from steel beams, 1 use	S.F.	C2	485	0.099	3.01	3.43		6.44	9.02
2 use	S.F.	C2	535	0.09	2.38	3.11		5.49	7.82
3 use	S.F.	C2	550	0.087	2.18	3.03		5.21	7.43
4 use	S.F.	C2	565	0.085	2.07	2.94		5.01	7.19
Floor slab, with 1-way joist pans, 1 use	S.F.	C2	415	0.116	8.93	4		12.93	16.52
2 use	S.F.	C2	445	0.108	6.68	3.74		10.42	13.59
3 use	S.F.	C2	475	0.101	5.95	3.5		9.45	12.32
4 use	S.F.	C2	500	0.096	5.56	3.32		8.88	11.68
With 2-way waffle domes, 1 use	S.F.	C2	405	0.119	9.21	4.11		13.32	17
2 use	S.F.	C2	450	0.107	6.96	3.69		10.65	13.86
3 use	S.F.	C2	460	0.104	6.23	3.61		9.84	12.87
4 use	S.F.	C2	470	0.102	5.84	3.54		9.38	12.38
Box out for slab openings, over 16" deep, 1 use	SFCA	C2	190	0.253	3.91	8.75		12.66	18.91
2 use	SFCA	C2	240	0.2	2.16	6.95		9.11	13.95
Shallow slab box outs, to 10 S.F.	Ea.	C2	42	1.143	13.03	39.62		52.65	80.18
Over 10 S.F. (use perimeter)	L.F.	C2	600	0.08	1.74	2.77		4.51	6.53
Bulkhead forms for slab, with keyway, 1 use, 2 piece	L.F.	C2	500	0.096	1.93	3.32		5.25	7.68
3 piece (see also edge forms)	L.F.	C2	460	0.104	2.14	3.61		5.75	8.38
Slab bulkhead form, 4-1/2" high, exp metal, w/keyway & stakes	L.F.	C1	1200	0.027	1.02	0.92		1.94	2.65
5-1/2" high	L.F.	C1	1100	0.029	1.28	1		2.28	3.07
7-1/2" high	L.F.	C1	960	0.033	1.54	1.14		2.68	3.61
9-1/2" high	L.F.	C1	840	0.038	1.67	1.31		2.98	4.02
Curb forms, wood, 6" to 12" high, on elevated slabs, 1 use	SFCA	C1	180	0.178	1.92	6.12		8.04	12.3
2 use	SFCA	C1	205	0.156	1.06	5.35		6.41	10.12
3 use	SFCA	C1	220	0.145	0.77	5		5.77	9.17
4 use	SFCA	C1	225	0.142	0.63	4.89		5.52	8.87
Edge forms to 6" high, on elevated slab, 4 use	L.F.	C1	500	0.064	0.24	2.2		2.44	3.93
7" to 12" high, 1 use	SFCA	C1	162	0.198	1.46	6.79		8.25	12.93
2 use	SFCA	C1	198	0.162	0.81	5.56		6.37	10.15
3 use	SFCA	C1	222	0.144	0.58	4.96		5.54	8.92
4 use	SFCA	C1	350	0.091	0.24	3.15		3.39	5.51
Depressed area forms to 12" high, 4 use	L.F.	C1	300	0.107	0.86	3.67		4.53	7.07
12" to 24" high, 4 use	L.F.	C1	175	0.183	1.18	6.28		7.46	11.8
Perimeter deck and rail for elevated slabs, straight	L.F.	C1	90	0.356	13.42	12.25		25.67	35.14
Curved	L.F.	C1	65	0.492	18.47	16.93		35.4	48.57
Void forms, round plastic, 8" high x 3" diameter	Ea.	C1	450	0.071	2.27	2.45		4.72	6.56
4" diameter	Ea.	C1	425	0.075	2.57	2.59		5.16	7.15

6" diameter	Ea.	C1	400	0.08	4.59	2.75		7.34	9.64
8" diameter	Ea.	C1	375	0.085	7.58	2.93		10.51	13.27
FORMS IN PLACE, EQUIPMENT FOUNDATIONS									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
1 use	SFCA	C2	160	0.3	3.08	10.39		13.47	20.73
2 use	SFCA	C2	190	0.253	1.7	8.75		10.45	16.47
3 use	SFCA	C2	200	0.24	1.24	8.33		9.57	15.2
4 use	SFCA	C2	205	0.234	1.01	8.13		9.14	14.64
FORMS IN PLACE, FOOTINGS									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Continuous wall, plywood, 1 use	SFCA	C1	375	0.085	7.41	2.93		10.34	13.04
2 use	SFCA	C1	440	0.073	4.07	2.5		6.57	8.65
3 use	SFCA	C1	470	0.068	2.95	2.35		5.3	7.16
4 use	SFCA	C1	485	0.066	2.41	2.27		4.68	6.44
Dowel supports for footings or beams, 1 use	L.F.	C1	500	0.064	1.04	2.2		3.24	4.82
Integral starter wall, to 4" high, 1 use	L.F.	C1	400	0.08	1.07	2.75		3.82	5.77
Keyway, 4 use, tapered wood, 2" x 4"	L.F.	1 Carp	530	0.015	0.25	0.59		0.84	1.25
2" x 6"	L.F.	1 Carp	500	0.016	0.36	0.63		0.99	1.43
Tapered plastic.	L.F.	1 Carp	530	0.015	1.5	0.59		2.09	2.63
For keyway hung from supports, add	L.F.	1 Carp	150	0.053	1.04	2.09		3.13	4.62
Pile cap, square or rectangular, job-built plywood, 1 use	SFCA	C1	290	0.11	3.28	3.8		7.08	9.95
2 use	SFCA	C1	346	0.092	1.81	3.18		4.99	7.29
3 use	SFCA	C1	371	0.086	1.31	2.96		4.27	6.4
4 use	SFCA	C1	383	0.084	1.07	2.87		3.94	5.98
Triangular or hexagonal, 1 use	SFCA	C1	225	0.142	3.84	4.89		8.73	12.4
2 use	SFCA	C1	280	0.114	2.11	3.93		6.04	8.85
3 use	SFCA	C1	305	0.105	1.54	3.61		5.15	7.7
4 use	SFCA	C1	315	0.102	1.25	3.5		4.75	7.18
Spread footings, job-built lumber, 1 use	SFCA	C1	305	0.105	2.36	3.61		5.97	8.61
2 use	SFCA	C1	371	0.086	1.31	2.96		4.27	6.39
3 use	SFCA	C1	401	0.08	0.94	2.75		3.69	5.61
4 use	SFCA	C1	414	0.077	0.76	2.65		3.41	5.27
Supports for dowels, plinths or templates, 2' x 2' footing	Ea.	C1	25	1.28	6.85	44.25		51.1	81.15
4' x 4' footing	Ea.	C1	22	1.455	13.76	49.91		63.67	98.45
8' x 8' footing	Ea.	C1	20	1.6	27.51	55.05		82.56	121.9
12' x 12' footing	Ea.	C1	17	1.882	33.69	64.83		98.52	145.11
Plinths, job-built plywood, 1 use	SFCA	C1	250	0.128	3.49	4.4		7.89	11.2
4 use	SFCA	C1	270	0.119	1.15	4.07		5.22	8.05
FORMS IN PLACE, GAS STATION FORMS									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Curb fascia, with template, 12 ga. steel, left in place, 9" high	L.F.	1 Carp	50	0.16	15.83	6.28		22.11	27.8
Sign or light bases, 18" diameter, 9" high	Ea.	1 Carp	9	0.889	99.95	34.99		134.94	167.11
30" diameter, 13" high	Ea.	1 Carp	8	1	158.34	39.1		197.44	238.9
Island forms, 10' long, 9" high, 3'-6" wide	Ea.	C1	10	3.2	443.59	110.1		553.69	671.67
4' wide	Ea.	C1	9	3.556	454.82	122.45		577.27	709.09
20' long, 9" high, 4' wide	Ea.	C1	6	5.333	735.57	183.16		918.73	1114.17
5' wide	Ea.	C1	5	6.4	763.64	220.21		983.85	1207.55
FORMS IN PLACE, GRADE BEAM									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Job-built plywood, 1 use	SFCA	C2	530	0.091	3.34	3.14		6.48	8.92
2 use	SFCA	C2	580	0.083	1.83	2.86		4.69	6.8
3 use	SFCA	C2	600	0.08	1.34	2.77		4.11	6.09
4 use	SFCA	C2	605	0.079	1.08	2.75		3.83	5.77
FORMS IN PLACE, MAT FOUNDATION									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Job-built plywood, 1 use	SFCA	C2	290	0.166	3.29	5.71		9	13.19
2 use	SFCA	C2	310	0.155	1.29	5.35		6.64	10.38
3 use	SFCA	C2	330	0.145	0.83	5.03		5.86	9.3
4 use	SFCA	C2	350	0.137	0.76	4.74		5.5	8.76
FORMS IN PLACE, SLAB ON GRADE									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Bulkhead forms w/keyway, wood, 6" high, 1 use	L.F.	C1	510	0.063	1.18	2.16		3.34	4.9
2 uses	L.F.	C1	400	0.08	0.65	2.75		3.4	5.31
4 uses	L.F.	C1	350	0.091	0.38	3.15		3.53	5.68
Bulkhead form for slab, 4-1/2" high, exp metal, incl keyway & stakes	L.F.	C1	1200	0.027	1.02	0.92		1.94	2.65
5-1/2" high	L.F.	C1	1100	0.029	1.28	1		2.28	3.07
7-1/2" high	L.F.	C1	960	0.033	1.54	1.14		2.68	3.61
9-1/2" high	L.F.	C1	840	0.038	1.67	1.31		2.98	4.02
Curb forms, wood, 6" to 12" high, on grade, 1 use	SFCA	C1	215	0.149	2.13	5.12		7.25	10.89
2 use	SFCA	C1	250	0.128	1.18	4.4		5.58	8.65
3 use	SFCA	C1	265	0.121	0.85	4.16		5.01	7.88
4 use	SFCA	C1	275	0.116	0.7	4		4.7	7.45
Edge forms, wood, 4 use, on grade, to 6" high	L.F.	C1	600	0.053	0.33	1.83		2.16	3.42
7" to 12" high	SFCA	C1	435	0.074	0.77	2.53		3.3	5.07
Over 12"	SFCA	C1	350	0.091	1.04	3.15		4.19	6.4
For depressed slabs, 4 use, to 12" high	L.F.	C1	300	0.107	0.85	3.67		4.52	7.06
To 24" high	L.F.	C1	175	0.183	1.13	6.28		7.41	11.75
For slab blockouts, to 12" high, 1 use	L.F.	C1	200	0.16	0.92	5.51		6.43	10.17
To 24" high, 1 use	L.F.	C1	120	0.267	1.17	9.16		10.33	16.57
Plastic (extruded), to 6" high, multiple use, on grade	L.F.	C1	800	0.04	7.47	1.38		8.85	10.54
Screed, 24 ga. metal key joint, see Section 03 15 16.30									
Wood, incl. wood stakes, 1" x 3"	L.F.	C1	900	0.036	0.95	1.22		2.17	3.1
2" x 4"	L.F.	C1	900	0.036	0.94	1.22		2.16	3.07
Trench forms in floor, wood, 1 use	SFCA	C1	160	0.2	2.12	6.89		9.01	13.81
2 use	SFCA	C1	175	0.183	1.17	6.28		7.45	11.78
3 use	SFCA	C1	180	0.178	0.84	6.12		6.96	11.12
4 use	SFCA	C1	185	0.173	0.69	5.97		6.66	10.68
Void form, corrugated fiberboard, 4" x 12", 4' long	S.F.	C1	3000	0.011	3.85	0.37		4.22	4.84
6" x 12", 4' long	S.F.	C1	3000	0.011	4.6	0.37		4.97	5.67
1/4" thick hardboard protective cover for void form	S.F.	2 Carp	1500	0.011	0.76	0.42		1.18	1.53
FORMS IN PLACE, WALLS									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Box out for wall openings, to 16" thick, to 10 S.F.	Ea.	C2	24	2	29.76	69.46		99.22	147.82
Over 10 S.F. (use perimeter)	L.F.	C2	280	0.171	2.56	5.92		8.48	12.7
Brick shelf, 4" w, add to wall forms, use wall area above shelf									
1 use	SFCA	C2	240	0.2	2.77	6.95		9.72	14.62
2 use	SFCA	C2	275	0.175	1.53	6.02		7.55	11.75
4 use	SFCA	C2	300	0.16	1.11	5.56		6.67	10.48
Bulkhead, wood with keyway, 1 use, 2 piece	L.F.	C2	265	0.181	2.46	6.28		8.74	13.15
Bulkhead forms with keyway, 1 piece expanded metal, 8" wall	L.F.	C1	1000	0.032	1.54	1.1		2.64	3.53
10" wall	L.F.	C1	800	0.04	1.67	1.38		3.05	4.13

12" wall	L.F.	C1	525	0.061	2.01	2.1	4.11	5.71
Buttress, to 8' high, 1 use	SFCA	C2	350	0.137	4.75	4.74	9.49	13.14
2 use	SFCA	C2	430	0.112	2.61	3.87	6.48	9.3
3 use	SFCA	C2	460	0.104	1.91	3.61	5.52	8.12
4 use	SFCA	C2	480	0.1	1.57	3.47	5.04	7.48
Corbel or haunch, to 12" wide, add to wall forms, 1 use	L.F.	C2	150	0.32	2.67	11.06	13.73	21.41
2 use	L.F.	C2	170	0.282	1.47	9.78	11.25	17.93
3 use	L.F.	C2	175	0.274	1.07	9.52	10.59	17.03
4 use	L.F.	C2	180	0.267	0.86	9.21	10.07	16.33
Wall, job-built plywood, to 8' high, 1 use	SFCA	C2	370	0.13	3.1	4.5	7.6	10.91
2 use	SFCA	C2	435	0.11	1.97	3.82	5.79	8.55
3 use	SFCA	C2	495	0.097	1.44	3.35	4.79	7.18
4 use	SFCA	C2	505	0.095	1.17	3.29	4.46	6.79
Over 8' to 16' high, 1 use	SFCA	C2	280	0.171	3.43	5.92	9.35	13.64
2 use	SFCA	C2	345	0.139	1.49	4.82	6.31	9.68
3 use	SFCA	C2	375	0.128	1.07	4.43	5.5	8.59
4 use	SFCA	C2	395	0.122	0.88	4.21	5.09	7.97
Over 16' high, 1 use	SFCA	C2	235	0.204	3.04	7.05	10.09	15.13
2 use	SFCA	C2	290	0.166	1.67	5.71	7.38	11.41
3 use	SFCA	C2	315	0.152	1.21	5.3	6.51	10.14
4 use	SFCA	C2	330	0.145	0.99	5.03	6.02	9.48
Radial, smooth curved, job-built plywood, 1 use	SFCA	C2	245	0.196	2.86	6.79	9.65	14.46
2 use	SFCA	C2	300	0.16	1.57	5.56	7.13	10.99
3 use	SFCA	C2	325	0.148	1.15	5.11	6.26	9.8
4 use	SFCA	C2	335	0.143	0.93	4.96	5.89	9.3
Below grade, job-built plywood, 1 use	SFCA	C2	225	0.213	3.04	7.41	10.45	15.71
2 use	SFCA	C2	225	0.213	1.68	7.41	9.09	14.2
3 use	SFCA	C2	225	0.213	1.39	7.41	8.8	13.89
4 use	SFCA	C2	225	0.213	1	7.41	8.41	13.44
Curved, 2' chords, job-built plywood, to 8' high, 1 use	SFCA	C2	290	0.166	2.41	5.71	8.12	12.22
2 use	SFCA	C2	355	0.135	1.33	4.68	6.01	9.28
3 use	SFCA	C2	385	0.125	0.97	4.32	5.29	8.27
4 use	SFCA	C2	400	0.12	0.79	4.16	4.95	7.81
Over 8' to 16' high, 1 use	SFCA	C2	290	0.166	1.03	5.71	6.74	10.72
2 use	SFCA	C2	355	0.135	0.57	4.68	5.25	8.45
3 use	SFCA	C2	385	0.125	0.42	4.32	4.74	7.66
4 use	SFCA	C2	400	0.12	0.34	4.16	4.5	7.33
Retaining wall, battered, job-built plyw'd, to 8' high, 1 use	SFCA	C2	300	0.16	2.28	5.56	7.84	11.76
2 use	SFCA	C2	355	0.135	1.25	4.68	5.93	9.2
3 use	SFCA	C2	375	0.128	0.91	4.43	5.34	8.41
4 use	SFCA	C2	390	0.123	0.74	4.26	5	7.91
Over 8' to 16' high, 1 use	SFCA	C2	240	0.2	2.48	6.95	9.43	14.31
2 use	SFCA	C2	295	0.163	1.37	5.66	7.03	10.92
3 use	SFCA	C2	305	0.157	0.99	5.45	6.44	10.2
4 use	SFCA	C2	320	0.15	0.81	5.2	6.01	9.53
Retaining wall form, plywood, smooth curve, 1 use	SFCA	C2	200	0.24	3.65	8.33	11.98	17.86
2 use	SFCA	C2	235	0.204	2.01	7.05	9.06	13.99
3 use	SFCA	C2	250	0.192	1.46	6.64	8.1	12.72
4 use	SFCA	C2	260	0.185	1.2	6.38	7.58	11.96
For gang wall forming, 192 S.F. sections, deduct	SFCA				10.00%	10.00%		
384 S.F. sections, deduct	SFCA				20.00%	20.00%		
Lintel or sill forms, 1 use	SFCA	1 Carp	30	0.267	3.65	10.44	14.09	21.35
2 use	SFCA	1 Carp	34	0.235	2.01	9.21	11.22	17.48
3 use	SFCA	1 Carp	36	0.222	1.46	8.7	10.16	16.07
4 use	SFCA	1 Carp	37	0.216	1.18	8.44	9.62	15.35
Modular prefabricated plywood, based on 20 uses of purchased forms, and 4 uses of bracing lumber								
To 8' high	SFCA	C2	800	0.06	1.22	2.08	3.3	4.82
Over 8' to 16' high	SFCA	C2	600	0.08	1.29	2.77	4.06	6.03
Plasters, 1 use	SFCA	C2	270	0.178	3.59	6.17	9.76	14.24
2 use	SFCA	C2	330	0.145	1.98	5.03	7.01	10.57
3 use	SFCA	C2	370	0.13	1.44	4.5	5.94	9.09
4 use	SFCA	C2	385	0.125	1.17	4.32	5.49	8.49
Steel framed plywood, based on 50 uses of purchased forms, and 4 uses of bracing lumber								
To 8' high	SFCA	C2	600	0.08	0.82	2.77	3.59	5.52
Over 8' to 16' high	SFCA	C2	450	0.107	0.82	3.69	4.51	7.07
Over 16' to 20' high	SFCA	C2	400	0.12	0.82	4.16	4.98	7.85
For elevated walls, add						10.00%		
For battered walls, 1 side battered, add					10.00%	10.00%		
For battered walls, 2 sides battered, add					15.00%	15.00%		

Architectural Cast-in-Place Concrete Forming
CONCRETE FORM LINERS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Liners for forms (add to wall forms), ABS plastic									
Aged wood, 4" wide, 1 use	SFCA	1 Carp	256	0.031	3.9	1.22		5.12	6.33
2 use	SFCA	1 Carp	256	0.031	2.14	1.22		3.36	4.4
3 use	SFCA	1 Carp	256	0.031	1.56	1.22		2.78	3.76
4 use	SFCA	1 Carp	256	0.031	1.27	1.22		2.49	3.43
Fractured rope rib, 1 use	SFCA	1 Carp	192	0.042	5.67	1.63		7.3	8.94
2 use	SFCA	1 Carp	192	0.042	3.11	1.63		4.74	6.12
3 use	SFCA	1 Carp	192	0.042	2.26	1.63		3.89	5.19
4 use	SFCA	1 Carp	192	0.042	1.83	1.63		3.46	4.73
Ribbed, 3/4" deep x 1-1/2" OC, 1 use	SFCA	1 Carp	224	0.036	5.67	1.4		7.07	8.56
2 use	SFCA	1 Carp	224	0.036	3.11	1.4		4.51	5.74
3 use	SFCA	1 Carp	224	0.036	2.26	1.4		3.66	4.81
4 use	SFCA	1 Carp	224	0.036	1.83	1.4		3.23	4.35
Rustic brick pattern, 1 use	SFCA	1 Carp	224	0.036	3.9	1.4		5.3	6.62
2 use	SFCA	1 Carp	224	0.036	2.14	1.4		3.54	4.69
3 use	SFCA	1 Carp	224	0.036	1.56	1.4		2.96	4.05
4 use	SFCA	1 Carp	224	0.036	1.27	1.4		2.67	3.72
3/8" striated, random, 1 use	SFCA	1 Carp	224	0.036	3.9	1.4		5.3	6.62
2 use	SFCA	1 Carp	224	0.036	2.14	1.4		3.54	4.69
3 use	SFCA	1 Carp	224	0.036	1.56	1.4		2.96	4.05
4 use	SFCA	1 Carp	224	0.036	1.27	1.4		2.67	3.72
Random vertical rustication, 1 use	SFCA	1 Carp	384	0.021	7.41	0.81		8.22	9.5
2 use	SFCA	1 Carp	384	0.021	4.07	0.81		4.88	5.84
3 use	SFCA	1 Carp	384	0.021	2.96	0.81		3.77	4.62
4 use	SFCA	1 Carp	384	0.021	2.4	0.81		3.21	4.01
Wood, beveled edge, 3/4" deep, 1 use	L.F.	1 Carp	384	0.021	0.18	0.81		0.99	1.56
1" deep, 1 use	L.F.	1 Carp	384	0.021	0.33	0.81		1.14	1.72

4" wide aged cedar, 1 use	SFCA	1 Carp	256	0.031	3.9	1.22	5.12	6.33
4" variable depth rough cedar	SFCA	1 Carp	224	0.036	5.67	1.4	7.07	8.56

Insulating Concrete Forming

INSULATING FORMS, LEFT IN PLACE

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
S.F. is for exterior face, but includes forms for both faces (total R22)									
4" wall, straight block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	90	0.178	23.58	6.95		30.53	37.97
90 corner block, exterior 16" x 38" x 22" (6.67 S.F.)	Ea.	2 Carp	75	0.213	28.64	8.33		36.97	45.33
45 corner block, exterior 16" x 34" x 18" (5.78 S.F.)	Ea.	2 Carp	75	0.213	28.08	8.33		36.41	44.77
6" wall, straight block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	90	0.178	24.71	6.95		31.66	38.53
90 corner block, exterior 16" x 32" x 24" (6.22 S.F.)	Ea.	2 Carp	75	0.213	28.08	8.33		36.41	44.77
45 corner block, exterior 16" x 26" x 18" (4.89 S.F.)	Ea.	2 Carp	75	0.213	27.51	8.33		35.84	44.21
Brick ledge block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	80	0.2	30.32	7.82		38.14	46.71
Taper top block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	80	0.2	28.64	7.82		36.46	44.46
8" wall, straight block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	90	0.178	25.83	6.95		32.78	40.22
90 corner block, exterior 16" x 34" x 26" (6.67 S.F.)	Ea.	2 Carp	75	0.213	34.81	8.33		43.14	52.07
45 corner block, exterior 16" x 28" x 20" (5.33 S.F.)	Ea.	2 Carp	75	0.213	28.64	8.33		36.97	45.33
Brick ledge block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	80	0.2	31.44	7.82		39.26	47.83
Taper top block, 16" x 48" (5.33 S.F.)	Ea.	2 Carp	80	0.2	29.76	7.82		37.58	45.59

ROOF DECK FORM BOARDS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes bulb tee sub-purlins @ 32-5/8" OC									
Non-asbestos fiber cement, 5/16" thick	S.F.	C13	2950	0.008	3.34	0.33	0.03	3.7	4.28
Fiberglass, 1" thick	S.F.	C13	2700	0.009	4.1	0.36	0.04	4.5	5.17
Wood fiber, 1" thick	S.F.	C13	2700	0.009	2.57	0.36	0.04	2.97	3.49

Permanent Stair Forming

FORMS IN PLACE, STAIRS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
(Slant length x width), 1 use	S.F.	C2	165	0.291	6.46	10.08		16.54	23.89
2 use	S.F.	C2	170	0.282	3.66	9.78		13.44	20.34
3 use	S.F.	C2	180	0.267	2.74	9.21		11.95	18.39
4 use	S.F.	C2	190	0.253	2.28	8.75		11.03	17.11
Alternate pricing method (1.0 L.F./S.F.), 1 use	LF Rsr	C2	100	0.48	6.46	16.62		23.08	34.85
2 use	LF Rsr	C2	105	0.457	3.66	15.85		19.51	30.27
3 use	LF Rsr	C2	110	0.436	2.74	15.13		17.87	28.22
4 use	LF Rsr	C2	115	0.417	2.28	14.46		16.74	26.68
Stairs, cast on sloping ground (length x width), 1 use	S.F.	C2	220	0.218	2.63	7.56		10.19	15.51
2 use	S.F.	C2	232	0.207	1.45	7.15		8.6	13.53
3 use	S.F.	C2	244	0.197	1.06	6.79		7.85	12.53
4 use	S.F.	C2	256	0.188	0.85	6.48		7.33	11.8

Concrete Accessories

Concrete Forming Accessories

CHAMFER STRIPS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Polyvinyl chloride, 1/2" wide with leg	L.F.	1 Carp	535	0.015	0.76	0.59		1.35	1.81
3/4" wide with leg	L.F.	1 Carp	525	0.015	0.84	0.6		1.44	1.92
1" radius with leg	L.F.	1 Carp	515	0.016	0.88	0.61		1.49	1.98
2" radius with leg	L.F.	1 Carp	500	0.016	1.68	0.63		2.31	2.89
Wood, 1/2" wide	L.F.	1 Carp	535	0.015	0.16	0.59		0.75	1.14
3/4" wide	L.F.	1 Carp	525	0.015	0.18	0.6		0.78	1.19
1" wide	L.F.	1 Carp	515	0.016	0.33	0.61		0.94	1.37

COLUMN FORM ACCESSORIES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Column clamps, adjustable to 24" x 24", buy	Set				193.16			193.16	212.25
Rent per month	Set				13.76			13.76	15.16
For sizes to 30" x 30", buy	Set				225.72			225.72	248.18
Rent per month	Set				15.78			15.78	17.41
For sizes to 36" x 36", buy	Set				278.5			278.5	306.58
Rent per month	Set				19.32			19.32	21.28
Bar type with wedges, 36" x 36", buy	Set				179.68			179.68	197.65
Rent per month	Set				12.58			12.58	13.81
48" x 48", buy	Set				247.06			247.06	271.77
Rent per month	Set				17.29			17.29	19.03
Scissor type with wedges, 36" x 36", buy	Set				154.97			154.97	169.57
Rent per month	Set				15.44			15.44	17.01
60" x 60", buy	Set				215.62			215.62	236.95
Rent per month	Set				21.62			21.62	23.58
Friction collars 2'-6" diam., buy	Set				3032.1			3032.1	3340.93
Rent per month	Set				212.25			212.25	233.58
4'-0" diam., buy	Set				3705.9			3705.9	4070.88
Rent per month	Set				259.41			259.41	285.24

HANGERS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Slab and beam form									
Banding iron									
3/4" x 22 ga., 14 L.F./lb. or 1/2" x 14 ga., 7 L.F./lb.	Lb.				1.54			1.54	1.7
Fascia ties, coil type, to 24" long	C				522.2			522.2	578.35
Frame ties to 8-1/8"	C				612.04			612.04	673.8
8-1/8" to 10-1/8"	C				645.73			645.73	713.11
10-1/8" to 12-1/8"	C				662.57			662.57	729.95
12-1/8" to 14-1/8"	C				685.03			685.03	752.41
14-1/8" to 16-1/8"	C				707.49			707.49	780.49
Half hanger	C				898.4			898.4	988.24
Haunch hanger, for 1" haunch									
Flange to 8-1/8"	C				668.19			668.19	735.57
8-1/8" to 10-1/8"	C				707.49			707.49	780.49
10-1/8" to 12-1/8"	C				741.18			741.18	819.79
12-1/8" to 14-1/8"	C				769.26			769.26	842.25
14-1/8" to 16-1/8"	C				808.56			808.56	892.79
Haunch half hanger, for 1" haunch	C				471.66			471.66	522.2
Snap tie hanger, to 30" overall length, 3000#	C				522.2			522.2	578.35
To 36" overall length	C				589.58			589.58	651.34
To 48" overall length	C				724.34			724.34	797.33
Steel beam hanger									
Flange to 8-1/8"	C				612.04			612.04	673.8
8-1/8" to 10-1/8"	C				645.73			645.73	713.11
10-1/8" to 12-1/8"	C				662.57			662.57	729.95
12-1/8" to 14-1/8"	C				685.03			685.03	752.41
14-1/8" to 16-1/8"	C				707.49			707.49	780.49
Coil threaded rods, continuous, 1/2" diameter	L.F.				1.58			1.58	1.74
Tie hangers to 30" overall length, 4000#	C				595.19			595.19	651.34

To 36" overall length	C		662.57		662.57	724.34
To 48" overall length	C		797.33		797.33	875.94
Tie back hanger, up to 12-1/8" flange	C		1656.43		1656.43	1824.88
Wire beam saddles, to 18" overall length						
1-1/2" joist, 7 gauge	C		668.19		668.19	735.57
4 gauge	C		701.88		701.88	774.87
Wire, black annealed, 15 gauge	Cwt.		172.94		172.94	189.79
16 gauge	Cwt.		190.91		190.91	210

SHORES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Erect and strip, by hand, horizontal members									
Aluminum joists and stringers	Ea.	2 Carp	60	0.267		10.44		10.44	17.34
Steel, adjustable beams	Ea.	2 Carp	45	0.356		13.89		13.89	23.15
Wood joists	Ea.	2 Carp	50	0.32		12.5		12.5	20.58
Wood stringers	Ea.	2 Carp	30	0.533		21.09		21.09	34.47
Vertical members to 10' high	Ea.	2 Carp	55	0.291		11.37		11.37	18.93
To 13' high	Ea.	2 Carp	50	0.32		12.5		12.5	20.58
To 16' high	Ea.	2 Carp	45	0.356		13.89		13.89	23.15
Reshoring	S.F.	2 Carp	1400	0.011	0.69	0.44		1.13	1.49
Flying truss system	SFCA	C17D	9600	0.009		0.35	0.08	0.43	0.68
Horizontal, aluminum joists, 6-1/4" high x 5' to 21' span, buy	L.F.				18.25			18.25	20.05
Beams, 7-1/4" high x 4' to 30' span	L.F.				21.34			21.34	23.58
Horizontal, steel beam, W8x10, 7' span, buy	Ea.				68.5			68.5	75.8
10' span	Ea.				79.73			79.73	87.59
15' span	Ea.				137.01			137.01	151.61
20' span	Ea.				193.16			193.16	212.25
Steel stringer, W8x10, 4' to 16' span, buy	L.F.				7.97			7.97	8.82
Rent for job duration, aluminum joist @ 2' OC, per mo.	SF Flr.				0.46			0.46	0.51
Steel W8x10	SF Flr.				0.2			0.2	0.22
Steel adjustable	SF Flr.				0.2			0.2	0.22
#1 post shore, steel, 5'-7" to 9'-6" high, 10,000# cap., buy	Ea.				174.07			174.07	192.03
#2 post shore, 7'-3" to 12'-10" high, 7800# capacity	Ea.				202.14			202.14	222.35
#3 post shore, 8'-10" to 16'-1" high, 3800# capacity	Ea.				220.11			220.11	242.57
Frame shoring systems, steel, 12,000#/leg, buy									
Frame, 2' wide x 6' high	Ea.				129.15			129.15	142.62
X-brace	Ea.				22.46			22.46	24.71
Base plate	Ea.				19.88			19.88	21.84
Screw jack	Ea.				43.8			43.8	47.73
U-head, 8" x 8"	Ea.				24.71			24.71	26.95

SLEEVES AND CHASES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Plastic, 1 use, 12" long, 2" diameter	Ea.	1 Carp	100	0.08	2.12	3.13		5.25	7.54
4" diameter	Ea.	1 Carp	90	0.089	5.95	3.48		9.43	12.33
6" diameter	Ea.	1 Carp	75	0.107	10.44	4.17		14.61	18.46
12" diameter	Ea.	1 Carp	60	0.133	30.88	5.2		36.08	42.95
Sheet metal, 2" diameter	Ea.	1 Carp	100	0.08	1.67	3.13		4.8	7.04
4" diameter	Ea.	1 Carp	90	0.089	2.09	3.48		5.57	8.06
6" diameter	Ea.	1 Carp	75	0.107	2.09	4.17		6.26	9.25
12" diameter	Ea.	1 Carp	60	0.133	4.18	5.2		9.38	13.29
Steel pipe, 2" diameter	Ea.	1 Carp	100	0.08	3.35	3.13		6.48	8.88
4" diameter	Ea.	1 Carp	90	0.089	12.18	3.48		15.66	19.18
6" diameter	Ea.	1 Carp	75	0.107	39.31	4.17		43.48	50.19
12" diameter	Ea.	1 Carp	60	0.133	93.77	5.2		98.97	111.45

SNAP TIES, 8-1/4" L&W (Lumber and wedge)

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
2250 lb., w/flat washer, 8" wall	C				104.44			104.44	114.55
10" wall	C				151.61			151.61	167.33
12" wall	C				157.22			157.22	172.94
16" wall	C				174.07			174.07	192.03
18" wall	C				180.8			180.8	198.77
With plastic cone, 8" wall	C				92.09			92.09	101.07
10" wall	C				95.46			95.46	105
12" wall	C				103.32			103.32	113.42
16" wall	C				113.42			113.42	124.65
18" wall	C				116.79			116.79	128.02
3350 lb., w/flat washer, 8" wall	C				188.66			188.66	207.76
10" wall	C				206.63			206.63	226.85
12" wall	C				211.12			211.12	232.46
16" wall	C				242.57			242.57	267.27
18" wall	C				252.68			252.68	278.5
With plastic cone, 8" wall	C				152.73			152.73	168.45
10" wall	C				167.33			167.33	184.17
12" wall	C				171.82			171.82	188.66
16" wall	C				196.53			196.53	216.74
18" wall	C				203.26			203.26	223.48

STAIR TREAD INSERTS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Cast nosing insert, abrasive surface, pre-drilled, includes screws									
Aluminum, 3" wide x 3' long	Ea.	1 Cefl	32	0.25	58.4	9.31		67.71	79.64
4' long	Ea.	1 Cefl	31	0.258	78.05	9.62		87.67	101.5
5' long	Ea.	1 Cefl	30	0.267	97.14	9.93		107.07	123.35
Extruded nosing insert, black abrasive strips, continuous anchor									
Aluminum, 3" wide x 3' long	Ea.	1 Cefl	64	0.125	37.62	4.65		42.27	49.11
4' long	Ea.	1 Cefl	60	0.133	50.54	4.96		55.5	63.62
5' long	Ea.	1 Cefl	56	0.143	62.89	5.3		68.19	77.7
Extruded nosing insert, black abrasive strips, pre-drilled, incl. screws									
Aluminum, 3" wide x 3' long	Ea.	1 Cefl	32	0.25	47.17	9.31		56.48	66.73
4' long	Ea.	1 Cefl	31	0.258	62.89	9.62		72.51	84.65
5' long	Ea.	1 Cefl	30	0.267	78.61	9.93		88.54	102.57

WALL AND FOUNDATION FORM ACCESSORIES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Coil tie system									
Coil ties 1/2", 6000 lb., to 8"	C				286.37			286.37	315.56
10" to 12"	C				342.52			342.52	381.82
18"	C				404.28			404.28	449.2
24"	C				482.89			482.89	533.43
36"	C				623.27			623.27	685.03
48"	C				763.64			763.64	842.25
3/4", 12,000 lb., to 8"	C				550.27			550.27	606.42
10" to 12"	C				617.65			617.65	679.42
18"	C				724.34			724.34	797.33

24"	C	819.79	819.79	904.02
36"	C	1123	1123	1235.3
48"	C	1431.83	1431.83	1600.28
1", 24,000 lb., to 8"	C	971.4	971.4	1066.85
10" to 12"	C	1151.08	1151.08	1263.38
18"	C	1319.53	1319.53	1431.83
24"	C	1656.43	1656.43	1824.88
36"	C	2161.78	2161.78	2386.38
48"	C	2667.13	2667.13	2919.8
1-1/4", 36,000 lb., to 8"	C	1431.83	1431.83	1572.2
10" to 12"	C	1740.65	1740.65	1909.1
18"	C	2217.93	2217.93	2442.53
24"	C	2723.28	2723.28	2975.95
36"	C	3705.9	3705.9	4070.88
48"	C	4688.53	4688.53	5137.73
Coil bolts, 1/2" diameter x 3" long	C	161.71	161.71	177.43
6" long	C	262.78	262.78	288.61
12" long	C	460.43	460.43	510.97
18" long	C	606.42	606.42	668.19
3/4" diameter x 3" long	C	353.75	353.75	387.44
6" long	C	651.34	651.34	718.72
12" long	C	971.4	971.4	1066.85
18" long	C	1263.38	1263.38	1403.75
1" diameter x 3" long	C	1516.05	1516.05	1656.43
6" long	C	1572.2	1572.2	1712.58
12" long	C	2386.38	2386.38	2610.98
18" long	C	3228.63	3228.63	3537.45
1-1/4" diameter x 3" long	C	2414.45	2414.45	2639.05
6" long	C	2442.53	2442.53	2695.2
12" long	C	3790.13	3790.13	4155.1
18" long	C	4856.98	4856.98	5334.25
Adjustable coil tie, 3/4" diameter, 20" long	C	1347.6	1347.6	1487.98
3/4" diameter, 36" long	C	1656.43	1656.43	1824.88
Tie cones, plastic, 1" setback length, 1/2" bolt diameter	C	97.7	97.7	107.25
3/4" bolt diameter	C	192.03	192.03	211.12
2" setback length, 1" bolt diameter	C	522.2	522.2	578.35
1-1/4" bolt diameter	C	628.88	628.88	696.26
Welding coil tie, 1/2" diameter x 4" long	C	263.91	263.91	290.86
3/4" diameter x 6" long	C	353.75	353.75	387.44
1" diameter x 8" long	C	550.27	550.27	606.42
Waler holders, 1/2" diameter	C	271.77	271.77	298.72
3/4" diameter	C	303.21	303.21	333.53
Flat washers, 4" x 5" x 1/4" for 3/4" diameter	C	1111.77	1111.77	1207.23
5" x 5" x 7/16" for 1" diameter	C	1487.98	1487.98	1628.35
Footings, turnbuckle form aligner	Ea.	18.7	18.7	20.55
Spreaders for footer, adjustable	Ea.	28.08	28.08	30.88
Lagstud, threaded, 1/2" diameter	C.L.F.	158.34	158.34	174.07
3/4" diameter	C.L.F.	336.9	336.9	370.59
1" diameter	C.L.F.	578.35	578.35	640.11
1-1/4" diameter	C.L.F.	1403.75	1403.75	1544.13
Lagnuts, 1/2" diameter	C	38.18	38.18	42.11
3/4" diameter	C	92.09	92.09	101.07
1" diameter	C	266.15	266.15	293.1
1-1/4" diameter	C	449.2	449.2	494.12
Lagnuts with handle, 1/2" diameter	C	348.13	348.13	381.82
3/4" diameter	C	527.81	527.81	583.96
1" diameter	C	977.01	977.01	1078.08
Plastic set back plugs for 3/4" diameter	C	111.18	111.18	122.41
Rock anchors, 1/2" diameter	C	1319.53	1319.53	1459.9
3/4" diameter	C	1768.73	1768.73	1937.18
1" diameter	C	1881.03	1881.03	2077.55
Batter washer, 1/2" diameter	C	746.8	746.8	819.79
Form oil, up to 1200 S.F./gallon coverage	Gal.	15.89	15.89	17.46
Up to 800 S.F./gallon	Gal.	24.14	24.14	26.39
Form patches, 1-3/4" diameter	C	31.44	31.44	34.81
2-3/4" diameter	C	53.9	53.9	59.52
Nail stakes, 3/4" diameter, 18" long	Ea.	2.22	2.22	2.45
24" long	Ea.	2.81	2.81	3.09
30" long	Ea.	3.55	3.55	3.91
36" long	Ea.	4.48	4.48	4.93
Pencil rods, 1/4" diameter	C.L.F.	55.03	55.03	60.64
Clamps for 1/4" pencil rods	Ea.	6.74	6.74	7.41
Clamping jacks for 1/4" pencil rod clamp	Ea.	88.16	88.16	96.58
She-bolts, 7/8" x 20"	C	5165.8	5165.8	5671.15
1-1/4" x 20"	C	6878.38	6878.38	7580.25
1-1/2" x 20"	C	9180.53	9180.53	10078.93
Inside rods, threaded, 1/2" diameter x 6"	C	213.37	213.37	234.71
1/2" diameter x 12"	C	309.95	309.95	342.52
5/8" diameter x 6"	C	415.51	415.51	454.82
5/8" diameter x 12"	C	606.42	606.42	662.57
3/4" diameter x 6"	C	460.43	460.43	505.35
3/4" diameter x 12"	C	640.11	640.11	707.49
For wing nuts, see taper ties				
Taper tie system				
Taper ties, 3/4" to 1/2" diameter x 30"	Ea.	63.45	63.45	69.63
1" to 3/4" diameter x 30"	Ea.	72.43	72.43	79.73
1-1/4" to 1" diameter x 30"	Ea.	84.23	84.23	92.65
Wing nuts, 1/2" diameter	C	780.49	780.49	859.1
3/4" diameter	C	836.64	836.64	920.86
7/8" diameter	C	915.25	915.25	1010.7
1" diameter	C	915.25	915.25	1010.7
1-1/8" diameter	C	1055.62	1055.62	1151.08
1-1/4" diameter	C	1179.15	1179.15	1291.45
1-1/2" diameter	C	1375.68	1375.68	1516.05
Flat washers, 1/4" x 3" x 4", for 1/2" diam. bolt	C	316.69	316.69	348.13
1/4" x 4" x 5", for 3/4" diam. bolt	C	1111.77	1111.77	1207.23
7/16" x 5" x 5", for 1" diam. bolt	C	1487.98	1487.98	1628.35
7/16" x 5" x 5", for 1-1/4" diam. bolt	C	1684.5	1684.5	1824.88
Wood form accessories				
Tie plate	C	701.88	701.88	774.87
Corner washer	C	1403.75	1403.75	1544.13

Panel bolt	C	1044.39	1044.39	1151.08
Panel wedge	C	184.17	184.17	202.14
Stud clamp	C	786.1	786.1	864.71

Waterstops

WATERSTOPS, PVC and Rubber

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
PVC, ribbed 3/16" thick, 4" wide	L.F.	1 Carp	155	0.052	1.61	2.02		3.63	5.11
6" wide	L.F.	1 Carp	145	0.055	2.71	2.16		4.87	6.57
With center bulb, 6" wide, 3/16" thick	L.F.	1 Carp	135	0.059	2.73	2.32		5.05	6.86
3/8" thick	L.F.	1 Carp	130	0.062	4.97	2.41		7.38	9.47
9" wide x 3/8" thick	L.F.	1 Carp	125	0.064	8.09	2.5		10.59	13.09
Dumbbell type, 6" wide, 3/16" thick	L.F.	1 Carp	150	0.053	3.87	2.09		5.96	7.74
3/8" thick	L.F.	1 Carp	145	0.055	4.17	2.16		6.33	8.17
9" wide, 3/8" thick, plain	L.F.	1 Carp	130	0.062	6.74	2.41		9.15	11.41
Center bulb	L.F.	1 Carp	130	0.062	9.55	2.41		11.96	14.5
Ribbed type, split, 3/16" thick, 6" wide	L.F.	1 Carp	145	0.055	2.25	2.16		4.41	6.06
3/8" thick	L.F.	1 Carp	130	0.062	5.23	2.41		7.64	9.78
Rubber, flat dumbbell, 3/8" thick, 6" wide	L.F.	1 Carp	145	0.055	5.11	2.16		7.27	9.21
9" wide	L.F.	1 Carp	135	0.059	11.12	2.32		13.44	16.1
Flat dumbbell split, 3/8" thick, 6" wide	L.F.	1 Carp	145	0.055	2.25	2.16		4.41	6.06
9" wide	L.F.	1 Carp	135	0.059	5.23	2.32		7.55	9.64
Center bulb, 1/4" thick, 6" wide	L.F.	1 Carp	145	0.055	7.75	2.16		9.91	12.12
9" wide	L.F.	1 Carp	135	0.059	15.44	2.32		17.76	20.87
Center bulb split, 3/8" thick, 6" wide	L.F.	1 Carp	145	0.055	6.4	2.16		8.56	10.61
9" wide	L.F.	1 Carp	135	0.059	11.06	2.32		13.38	16.04

Waterstop fittings, rubber, flat

Dumbbell or center bulb, 3/8" thick,

Field union, 6" wide	Ea.	1 Carp	50	0.16	35.37	6.28		41.65	49.7
9" wide	Ea.	1 Carp	50	0.16	39.31	6.28		45.59	53.63
Flat cross, 6" wide	Ea.	1 Carp	30	0.267	55.59	10.44		66.03	78.54
9" wide	Ea.	1 Carp	30	0.267	75.8	10.44		86.24	101
Flat tee, 6" wide	Ea.	1 Carp	30	0.267	53.34	10.44		63.78	76.3
9" wide	Ea.	1 Carp	30	0.267	70.19	10.44		80.63	94.27
Flat ell, 6" wide	Ea.	1 Carp	40	0.2	52.22	7.82		60.04	70.85
9" wide	Ea.	1 Carp	40	0.2	64.57	7.82		72.39	84.33
Vertical tee, 6" wide	Ea.	1 Carp	25	0.32	23.02	12.5		35.52	46.41
9" wide	Ea.	1 Carp	25	0.32	32.01	12.5		44.51	55.39
Vertical ell, 6" wide	Ea.	1 Carp	35	0.229	23.02	8.95		31.97	40.14
9" wide	Ea.	1 Carp	35	0.229	44.36	8.95		53.31	63.72

Concrete Construction Joints

CONTROL JOINTS, SAW CUT

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Sawcut control joints in green concrete									
1" depth	L.F.	C27	2000	0.008	0.03	0.3	0.05	0.38	0.58
1-1/2" depth	L.F.	C27	1800	0.009	0.06	0.33	0.06	0.45	0.67
2" depth	L.F.	C27	1600	0.01	0.08	0.37	0.06	0.51	0.77
Sawcut joint reservoir in cured concrete									
3/8" wide x 3/4" deep, with single saw blade	L.F.	C27	1000	0.016	0.06	0.6	0.1	0.76	1.15
1/2" wide x 1" deep, with double saw blades	L.F.	C27	900	0.018	0.11	0.66	0.11	0.88	1.32
3/4" wide x 1-1/2" deep, with double saw blades	L.F.	C27	800	0.02	0.24	0.74	0.13	1.11	1.61
Water blast joint to wash away laitance, 2 passes	L.F.	C29	2500	0.003		0.09	0.03	0.12	0.19
Air blast joint to blow out debris and air dry, 2 passes	L.F.	C28	2000	0.004		0.14	0.01	0.15	0.25

For backer rod, see Section 07 91 23.10

For joint sealant, see Section 03 15 16.30 or 07 92 13.20

For joint sealant, see Section 07 92 13.20

For joint sealant, see Section 03 15 16.30

For replacement of joint sealant, see Section 07 01 90.81

EXPANSION JOINTS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Keyed, cold, 24 ga., incl. stakes, 3-1/2" high	L.F.	1 Carp	200	0.04	0.95	1.56		2.51	3.66
4-1/2" high	L.F.	1 Carp	200	0.04	1.02	1.56		2.58	3.72
5-1/2" high	L.F.	1 Carp	195	0.041	1.28	1.61		2.89	4.07
7-1/2" high	L.F.	1 Carp	190	0.042	1.54	1.65		3.19	4.44
9-1/2" high	L.F.	1 Carp	185	0.043	1.67	1.69		3.36	4.65
Poured asphalt, plain, 1/2" x 1"	L.F.	1 Clab	450	0.018	0.27	0.54		0.81	1.19
1" x 2"	L.F.	1 Clab	400	0.02	1.07	0.61		1.68	2.19
Neoprene, liquid, cold applied, 1/2" x 1"	L.F.	1 Clab	450	0.018	2.56	0.54		3.1	3.72
1" x 2"	L.F.	1 Clab	400	0.02	10.22	0.61		10.83	12.3
Polyurethane, poured, 2 part, 1/2" x 1"	L.F.	1 Clab	400	0.02	1.56	0.61		2.17	2.73
1" x 2"	L.F.	1 Clab	350	0.023	6.23	0.69		6.92	8
Rubberized asphalt, hot or cold applied, 1/2" x 1"	L.F.	1 Clab	450	0.018	0.34	0.54		0.88	1.27
1" x 2"	L.F.	1 Clab	400	0.02	1.34	0.61		1.95	2.48
Hot applied, fuel resistant, 1/2" x 1"	L.F.	1 Clab	450	0.018	0.51	0.54		1.05	1.46
1" x 2"	L.F.	1 Clab	400	0.02	2.01	0.61		2.62	3.21
Premolded, bituminous fiber, 1/2" x 6"	L.F.	1 Carp	375	0.021	0.47	0.83		1.3	1.91
1" x 12"	L.F.	1 Carp	300	0.027	2.21	1.04		3.25	4.18
Concrete expansion joint, recycled paper and fiber, 1/2" x 6"	L.F.	1 Carp	390	0.021	0.47	0.8		1.27	1.86
1/2" x 12"	L.F.	1 Carp	360	0.022	0.94	0.86		1.8	2.48
Cork with resin binder, 1/2" x 6"	L.F.	1 Carp	375	0.021	1.31	0.83		2.14	2.84
1" x 12"	L.F.	1 Carp	300	0.027	3.59	1.04		4.63	5.69
Neoprene sponge, closed cell, 1/2" x 6"	L.F.	1 Carp	375	0.021	2.59	0.83		3.42	4.24
1" x 12"	L.F.	1 Carp	300	0.027	9.94	1.04		10.98	12.63
Polyethylene foam, 1/2" x 6"	L.F.	1 Carp	375	0.021	0.58	0.83		1.41	2.03
1" x 12"	L.F.	1 Carp	300	0.027	3.12	1.04		4.16	5.18
Polyethylene backer rod, 3/8" diameter	L.F.	1 Carp	460	0.017	0.03	0.68		0.71	1.17
3/4" diameter	L.F.	1 Carp	460	0.017	0.08	0.68		0.76	1.21
1" diameter	L.F.	1 Carp	460	0.017	0.13	0.68		0.81	1.28
Polyurethane foam, with polybutylene, 1/2" x 1/2"	L.F.	1 Carp	475	0.017	1.34	0.66		2	2.56
1" x 1"	L.F.	1 Carp	450	0.018	3.4	0.7		4.1	4.89
Polyurethane foam, regular, closed cell, 1/2" x 6"	L.F.	1 Carp	375	0.021	1	0.83		1.83	2.49
1" x 12"	L.F.	1 Carp	300	0.027	3.54	1.04		4.58	5.64
Polyvinyl chloride foam, closed cell, 1/2" x 6"	L.F.	1 Carp	375	0.021	2.7	0.83		3.53	4.35
1" x 12"	L.F.	1 Carp	300	0.027	9.26	1.04		10.3	11.96
Rod, 3/8" polyethylene/rubberized asphalt sealer, 1/4" x 3/4"	L.F.	2 Clab	600	0.027	0.2	0.81		1.01	1.57
Rubber, gray sponge, 1/2" x 6"	L.F.	1 Carp	375	0.021	2.27	0.83		3.1	3.88
1" x 12"	L.F.	1 Carp	300	0.027	8.14	1.04		9.18	10.72
Redwood heartwood, 1" x 4"	L.F.	1 Carp	400	0.02	1.3	0.78		2.08	2.74
1" x 6"	L.F.	1 Carp	375	0.021	1.97	0.83		2.8	3.56
For installation in walls, add						75.00%			
For installation in boxouts, add						25.00%			

Cast-In Concrete Anchors

ANCHOR BOLT ACCESSORIES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
For anchor bolts set in fresh concrete, see Section 03 15 19.10									
Anchor bolt sleeve, plastic, 1" diameter bolts	Ea.	1 Carp	60	0.133	15.05	5.2		20.25	25.26
1-1/2" diameter	Ea.	1 Carp	28	0.286	22.18	11.16		33.34	42.71
2" diameter	Ea.	1 Carp	24	0.333	21.84	13.02		34.86	45.75
3" diameter	Ea.	1 Carp	20	0.4	39.87	15.64		55.51	70.04
Templates, steel, 8" bolt spacing	Ea.	2 Carp	16	1	12.3	39.1		51.4	78.36
12" bolt spacing	Ea.	2 Carp	15	1.067	12.86	41.67		54.53	83.55
16" bolt spacing	Ea.	2 Carp	14	1.143	15.39	44.76		60.15	91.05
24" bolt spacing	Ea.	2 Carp	12	1.333	20.55	51.96		72.51	109.41
Wood, 8" bolt spacing	Ea.	2 Carp	16	1	0.94	39.1		40.04	65.87
12" bolt spacing	Ea.	2 Carp	15	1.067	1.25	41.67		42.92	70.83
16" bolt spacing	Ea.	2 Carp	14	1.143	1.56	44.76		46.32	75.81
24" bolt spacing	Ea.	2 Carp	16	1	2.19	39.1		41.29	67.23

ANCHOR BOLTS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Made from recycled materials									
Single bolts installed in fresh concrete, no templates									
Hooked w/nut and washer, 1/2" diameter, 8" long	Ea.	1 Carp	132	0.061	1.59	2.37		3.96	5.69
12" long	Ea.	1 Carp	131	0.061	1.77	2.39		4.16	5.92
5/8" diameter, 8" long	Ea.	1 Carp	129	0.062	4.27	2.43		6.7	8.72
12" long	Ea.	1 Carp	127	0.063	5.24	2.46		7.7	9.88
3/4" diameter, 8" long	Ea.	1 Carp	127	0.063	5.24	2.46		7.7	9.88
12" long	Ea.	1 Carp	125	0.064	6.57	2.5		9.07	11.35
2-bolt pattern, including job-built 2-hole template, per set									
J-type, incl. hex nut & washer, 1/2" diameter x 6" long	Set	1 Carp	21	0.381	5.78	14.92		20.7	31.04
12" long	Set	1 Carp	21	0.381	6.46	14.92		21.38	31.83
18" long	Set	1 Carp	21	0.381	7.52	14.92		22.44	33.01
3/4" diameter x 8" long	Set	1 Carp	20	0.4	13.42	15.64		29.06	41.01
12" long	Set	1 Carp	20	0.4	16.06	15.64		31.7	43.87
18" long	Set	1 Carp	20	0.4	19.99	15.64		35.63	48.25
1" diameter x 12" long	Set	1 Carp	19	0.421	25.27	16.46		41.73	55.35
18" long	Set	1 Carp	19	0.421	30.32	16.46		46.78	60.4
24" long	Set	1 Carp	19	0.421	35.94	16.46		52.4	67.14
36" long	Set	1 Carp	18	0.444	48.29	17.39		65.68	82.15
1-1/2" diameter x 18" long	Set	1 Carp	17	0.471	46.6	18.42		65.02	81.46
24" long	Set	1 Carp	16	0.5	54.47	19.55		74.02	92.49
L-type, incl. hex nut & washer, 3/4" diameter x 12" long	Set	1 Carp	20	0.4	15.83	15.64		31.47	43.65
18" long	Set	1 Carp	20	0.4	19.48	15.64		35.12	47.63
24" long	Set	1 Carp	20	0.4	23.02	15.64		38.66	51.51
30" long	Set	1 Carp	20	0.4	28.64	15.64		44.28	57.68
36" long	Set	1 Carp	20	0.4	32.01	15.64		47.65	61.61
1" diameter x 12" long	Set	1 Carp	19	0.421	24.71	16.46		41.17	54.22
18" long	Set	1 Carp	19	0.421	29.76	16.46		46.22	59.84
24" long	Set	1 Carp	19	0.421	35.94	16.46		52.4	67.14
30" long	Set	1 Carp	19	0.421	42.11	16.46		58.57	73.31
36" long	Set	1 Carp	18	0.444	47.73	17.39		65.12	81.03
42" long	Set	1 Carp	18	0.444	57.27	17.39		74.66	91.7
48" long	Set	1 Carp	18	0.444	64.01	17.39		81.4	99
1-1/4" diameter x 18" long	Set	1 Carp	18	0.444	37.62	17.39		55.01	70.36
24" long	Set	1 Carp	18	0.444	43.8	17.39		61.19	77.1
30" long	Set	1 Carp	17	0.471	50.54	18.42		68.96	85.95
36" long	Set	1 Carp	17	0.471	56.71	18.42		75.13	93.25
42" long	Set	2 Carp	32	0.5	64.01	19.55		83.56	102.6
48" long	Set	2 Carp	32	0.5	72.43	19.55		91.98	112.14
54" long	Set	2 Carp	31	0.516	85.35	20.17		105.52	127.21
60" long	Set	2 Carp	31	0.516	93.21	20.17		113.38	136.19
1-1/2" diameter x 18" long	Set	2 Carp	33	0.485	55.59	18.93		74.52	92.58
24" long	Set	2 Carp	32	0.5	64.01	19.55		83.56	103.16
30" long	Set	2 Carp	31	0.516	72.43	20.17		92.6	112.61
36" long	Set	2 Carp	30	0.533	82.54	21.09		103.63	125.43
42" long	Set	2 Carp	30	0.533	93.77	21.09		114.86	137.22
48" long	Set	2 Carp	29	0.552	105	21.61		126.61	151.69
54" long	Set	2 Carp	28	0.571	126.9	22.12		149.02	176.29
60" long	Set	2 Carp	28	0.571	139.25	22.12		161.37	189.77
1-3/4" diameter x 18" long	Set	2 Carp	31	0.516	71.87	20.17		92.04	112.61
24" long	Set	2 Carp	30	0.533	83.66	21.09		104.75	126.56
30" long	Set	2 Carp	29	0.552	97.14	21.61		118.75	142.71
36" long	Set	2 Carp	28	0.571	110.05	22.12		132.17	158.32
42" long	Set	2 Carp	27	0.593	123.53	23.15		146.68	174.47
48" long	Set	2 Carp	26	0.615	134.76	24.18		158.94	189.49
54" long	Set	2 Carp	26	0.615	167.33	24.18		191.51	224.3
60" long	Set	2 Carp	25	0.64	180.8	25.21		206.01	240.44
2" diameter x 24" long	Set	2 Carp	27	0.593	135.88	23.15		159.03	187.95
30" long	Set	2 Carp	27	0.593	152.73	23.15		175.88	207.04
36" long	Set	2 Carp	26	0.615	167.33	24.18		191.51	224.3
42" long	Set	2 Carp	25	0.64	186.42	25.21		211.63	247.18
48" long	Set	2 Carp	24	0.667	213.37	26.24		239.61	277.93
54" long	Set	2 Carp	23	0.696	253.8	27.27		281.07	324.91
60" long	Set	2 Carp	23	0.696	272.89	27.27		300.16	345.12
66" long	Set	2 Carp	22	0.727	291.98	28.3		320.28	368.51
72" long	Set	2 Carp	21	0.762	318.93	29.84		348.77	403.14
4-bolt pattern, including job-built 4-hole template, per set									
J-type, incl. hex nut & washer, 1/2" diameter x 6" long	Set	1 Carp	19	0.421	8.59	16.46		25.05	36.76
12" long	Set	1 Carp	19	0.421	10.05	16.46		26.51	38.33
18" long	Set	1 Carp	18	0.444	12.18	17.39		29.57	42.17
3/4" diameter x 8" long	Set	1 Carp	17	0.471	24.14	18.42		42.56	56.75
12" long	Set	1 Carp	17	0.471	29.2	18.42		47.62	62.37
18" long	Set	1 Carp	17	0.471	37.06	18.42		55.48	71.35
1" diameter x 12" long	Set	1 Carp	16	0.5	47.73	19.55		67.28	85.19
18" long	Set	1 Carp	15	0.533	57.27	21.09		78.36	97.36
24" long	Set	1 Carp	15	0.533	69.06	21.09		90.15	110.27
36" long	Set	1 Carp	15	0.533	93.77	21.09		114.86	137.22
1-1/2" diameter x 18" long	Set	1 Carp	13	0.615	89.84	24.18		114.02	138.95
24" long	Set	1 Carp	12	0.667	106.12	26.24		132.36	160.01
L-type, incl. hex nut & washer, 3/4" diameter x 12" long	Set	1 Carp	17	0.471	28.64	18.42		47.06	61.8
18" long	Set	1 Carp	17	0.471	35.94	18.42		54.36	70.23
24" long	Set	1 Carp	17	0.471	43.24	18.42		61.66	78.09
30" long	Set	1 Carp	16	0.5	53.9	19.55		73.45	91.93
36" long	Set	1 Carp	16	0.5	61.2	19.55		80.75	99.79

1" diameter x 12" long	Set	1 Carp	16	0.5	46.04	19.55	65.59	82.95
18" long	Set	1 Carp	15	0.533	56.71	21.09	77.8	96.8
24" long	Set	1 Carp	15	0.533	69.06	21.09	90.15	110.27
30" long	Set	1 Carp	15	0.533	80.86	21.09	101.95	123.75
36" long	Set	1 Carp	15	0.533	92.09	21.09	113.18	136.1
42" long	Set	1 Carp	14	0.571	111.74	22.12	133.86	159.45
48" long	Set	1 Carp	14	0.571	124.65	22.12	146.77	174.05
1-1/4" diameter x 18" long	Set	1 Carp	14	0.571	71.87	22.12	93.99	116.21
24" long	Set	1 Carp	14	0.571	85.35	22.12	107.47	130.81
30" long	Set	1 Carp	13	0.615	98.26	24.18	122.44	147.94
36" long	Set	1 Carp	13	0.615	111.18	24.18	135.36	162.54
42" long	Set	2 Carp	25	0.64	124.65	25.21	149.86	179.8
48" long	Set	2 Carp	24	0.667	142.62	26.24	168.86	199.32
54" long	Set	2 Carp	23	0.696	167.33	27.27	194.6	229.45
60" long	Set	2 Carp	23	0.696	183.05	27.27	210.32	247.42
1-1/2" diameter x 18" long	Set	2 Carp	25	0.64	107.81	25.21	133.02	160.71
24" long	Set	2 Carp	24	0.667	125.78	26.24	152.02	181.35
30" long	Set	2 Carp	23	0.696	141.5	27.27	168.77	201.38
36" long	Set	2 Carp	22	0.727	161.71	28.3	190.01	225.89
42" long	Set	2 Carp	22	0.727	184.17	28.3	212.47	250.59
48" long	Set	2 Carp	21	0.762	206.63	29.84	236.47	276.24
54" long	Set	2 Carp	20	0.8	251.55	31.38	282.93	328.22
60" long	Set	2 Carp	20	0.8	275.14	31.38	306.52	354.05
1-3/4" diameter x 18" long	Set	2 Carp	22	0.727	140.38	28.3	168.68	202.3
24" long	Set	2 Carp	21	0.762	163.96	29.84	193.8	230.19
30" long	Set	2 Carp	21	0.762	190.91	29.84	220.75	259.39
36" long	Set	2 Carp	20	0.8	217.86	31.38	249.24	291.16
42" long	Set	2 Carp	19	0.842	243.69	32.93	276.62	322.94
48" long	Set	2 Carp	18	0.889	267.27	34.99	302.26	351.85
54" long	Set	2 Carp	18	0.889	331.29	34.99	366.28	422.6
60" long	Set	2 Carp	17	0.941	359.36	37.04	396.4	454.28
2" diameter x 24" long	Set	2 Carp	19	0.842	268.4	32.93	301.33	349.89
30" long	Set	2 Carp	18	0.889	302.09	34.99	337.08	390.03
36" long	Set	2 Carp	18	0.889	331.29	34.99	366.28	422.6
42" long	Set	2 Carp	17	0.941	370.59	37.04	407.63	465.51
48" long	Set	2 Carp	16	1	426.74	39.1	465.84	530.88
54" long	Set	2 Carp	15	1.067	505.35	41.67	547.02	625.35
60" long	Set	2 Carp	15	1.067	544.66	41.67	586.33	664.65
66" long	Set	2 Carp	14	1.143	578.35	44.76	623.11	714.2
72" long	Set	2 Carp	14	1.143	634.5	44.76	679.26	770.35
For galvanized, add	Ea.				75.00%			

DOVETAIL ANCHOR SYSTEM

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Dovetail anchor slot, galvanized, foam-filled, 26 ga.	L.F.	1 Carp	425	0.019	1.31	0.74		2.05	2.67
24 ga.	L.F.	1 Carp	400	0.02	1.81	0.78		2.59	3.29
22 ga.	L.F.	1 Carp	400	0.02	1.97	0.78		2.75	3.47
Stainless steel, foam-filled, 26 ga.	L.F.	1 Carp	375	0.021	2.1	0.83		2.93	3.7
Dovetail brick anchor, corrugated, galvanized, 3-1/2" long, 16 ga.	C	1 Bric	10.5	0.762	45.48	28.3		73.78	97.3
12 ga.	C	1 Bric	10.5	0.762	67.38	28.3		95.68	121.45
Seismic, galvanized, 3-1/2" long, 16 ga.	C	1 Bric	10.5	0.762	79.73	28.3		108.03	134.92
12 ga.	C	1 Bric	10.5	0.762	94.33	28.3		122.63	151.21
Dovetail cavity wall, corrugated, galvanized, 5-1/2" long, 16 ga.	C	1 Bric	10.5	0.762	44.92	28.3		73.22	96.74
12 ga.	C	1 Bric	10.5	0.762	64.01	28.3		92.31	117.52
Dovetail furring anchors, corrugated, galvanized, 1-1/2" long, 16 ga.	C	1 Bric	10.5	0.762	26.95	28.3		55.25	77.09
12 ga.	C	1 Bric	10.5	0.762	34.81	28.3		63.11	85.51
Dovetail stone panel anchors, galvanized, 1/8" x 1" wide, 3-1/2" long	C	1 Bric	10.5	0.762	108.93	28.3		137.23	167.49
1/4" x 1" wide	C	1 Bric	10.5	0.762	184.17	28.3		212.47	249.47

INSERTS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Inserts, slotted nut type for 3/4" bolts, 4" long	Ea.	1 Carp	84	0.095	22.46	3.72		26.18	31.44
6" long	Ea.	1 Carp	84	0.095	25.83	3.72		29.55	34.25
8" long	Ea.	1 Carp	84	0.095	34.25	3.72		37.97	43.79
Slotted, strap type, 4" long	Ea.	1 Carp	84	0.095	24.71	3.72		28.43	33.12
6" long	Ea.	1 Carp	84	0.095	27.51	3.72		31.23	36.49
8" long	Ea.	1 Carp	84	0.095	36.5	3.72		40.22	46.6
Strap for slotted insert, 4" long	Ea.	1 Carp	84	0.095	13.36	3.72		17.08	20.88
6" long	Ea.	1 Carp	84	0.095	14.54	3.72		18.26	22.17
8" long	Ea.	1 Carp	84	0.095	18.14	3.72		21.86	26.1
10" long	Ea.	1 Carp	84	0.095	22.01	3.72		25.73	30.31

Thin slab, ferrule type

1/2" diameter bolt	Ea.	1 Carp	60	0.133	4.74	5.2		9.94	13.91
5/8" diameter bolt	Ea.	1 Carp	60	0.133	5.84	5.2		11.04	15.1
3/4" diameter bolt	Ea.	1 Carp	60	0.133	6.79	5.2		11.99	16.17
1" diameter bolt	Ea.	1 Carp	60	0.133	10.95	5.2		16.15	20.72

Loop ferrule type

1/4" diameter bolt	Ea.	1 Carp	84	0.095	2.61	3.72		6.33	9.03
3/8" diameter bolt	Ea.	1 Carp	84	0.095	2.61	3.72		6.33	9.03
1/2" diameter bolt	Ea.	1 Carp	84	0.095	3.12	3.72		6.84	9.61
5/8" diameter bolt	Ea.	1 Carp	84	0.095	4.18	3.72		7.9	10.76
3/4" diameter bolt	Ea.	1 Carp	84	0.095	4.69	3.72		8.41	11.34
7/8" diameter bolt	Ea.	1 Carp	84	0.095	12.13	3.72		15.85	19.48
1" diameter bolt	Ea.	1 Carp	84	0.095	12.13	3.72		15.85	19.48

Threaded coil type with filler plug

3/4" diameter bolt	Ea.	1 Carp	60	0.133	5.57	5.2		10.77	14.82
1" diameter bolt	Ea.	1 Carp	60	0.133	10.5	5.2		15.7	20.21
1-1/4" diameter bolt	Ea.	1 Carp	60	0.133	19.82	5.2		25.02	30.54
1-1/2" diameter bolt	Ea.	1 Carp	60	0.133	32.01	5.2		37.21	43.51

Wedge type

For 3/4" diameter bolt	Ea.	1 Carp	60	0.133	10.89	5.2		16.09	20.66
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Askew head bolts, black

3/4" diameter, 1-1/2" long	Ea.				5.13			5.13	5.67
2" long	Ea.				5.62			5.62	6.18
3" long	Ea.				10.22			10.22	11.23

Nuts, black

3/4" bolt	Ea.				2.13			2.13	2.35
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Cut washers, black

3/4" bolt	Ea.				1.36			1.36	1.49
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For galvanized inserts, add

Machinery Anchors

MACHINERY ANCHORS, heavy duty, incl. sleeve, floating base nut,

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
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Stud & bolt, 1/2" diameter	Ea.	E16	40	0.4	58.96	17.08	2.55	78.59	98.3
5/8" diameter	Ea.	E16	35	0.457	68.5	19.5	2.91	90.91	112.91
3/4" diameter	Ea.	E16	30	0.533	81.42	22.64	3.4	107.46	133.71
7/8" diameter	Ea.	E16	25	0.64	92.65	27.27	4.09	124.01	154.48
1" diameter	Ea.	E16	20	0.8	99.39	33.96	5.1	138.45	175.8
1-1/4" diameter	Ea.	E16	15	1.067	132.51	45.28	6.79	184.58	234.24

Reinforcement Bars

Reinforcing Steel Accessories

REBAR ACCESSORIES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Steel & plastic made from recycled materials									
Beam bolsters (BB), lower, 1-1/2" high, plain steel	C.L.F.				39.9			39.9	43.89
Galvanized	C.L.F.				47.88			47.88	52.44
Stainless tipped legs	C.L.F.				524.4			524.4	575.7
Plastic tipped legs	C.L.F.				58.14			58.14	63.84
Epoxy dipped	C.L.F.				100.32			100.32	110.58
2" high, plain	C.L.F.				50.16			50.16	55.29
Galvanized	C.L.F.				60.42			60.42	66.12
Stainless tipped legs	C.L.F.				535.8			535.8	587.1
Plastic tipped legs	C.L.F.				68.4			68.4	75.24
Epoxy dipped	C.L.F.				114			114	125.4
Upper (BBU), 1-1/2" high, plain steel	C.L.F.				96.9			96.9	106.59
3" high	C.L.F.				109.44			109.44	120.84
Slab bolsters, continuous (SB), 1" high, plain steel	C.L.F.				34.2			34.2	37.62
Galvanized	C.L.F.				41.04			41.04	45.03
Stainless tipped legs	C.L.F.				518.7			518.7	570
Plastic tipped legs	C.L.F.				50.16			50.16	55.29
2" high, plain steel	C.L.F.				43.32			43.32	47.88
Galvanized	C.L.F.				51.87			51.87	57
Stainless tipped legs	C.L.F.				530.1			530.1	581.4
Plastic tipped legs	C.L.F.				58.14			58.14	63.84
For bolsters with wire runners (SBR), add	C.L.F.				46.74			46.74	51.3
For bolsters with plates (SBP), add	C.L.F.				110.58			110.58	121.98
Bag ties, 16 ga., plain, 4" long	C				4.56			4.56	5.02
5" long	C				5.7			5.7	6.27
6" long	C				4.56			4.56	5.02
7" long	C				5.7			5.7	6.27
High chairs, individual (HC), 3" high, plain steel	C				70.68			70.68	77.52
Galvanized	C				84.93			84.93	93.48
Stainless tipped legs	C				552.9			552.9	609.9
Plastic tipped legs	C				76.38			76.38	83.79
5" high, plain	C				102.6			102.6	112.86
Galvanized	C				123.12			123.12	135.66
Stainless tipped legs	C				587.1			587.1	644.1
Plastic tipped legs	C				112.86			112.86	124.26
8" high, plain	C				150.48			150.48	165.3
Galvanized	C				180.12			180.12	198.36
Stainless tipped legs	C				632.7			632.7	701.1
Plastic tipped legs	C				166.44			166.44	183.54
12" high, plain	C				359.1			359.1	393.3
Galvanized	C				433.2			433.2	473.1
Stainless tipped legs	C				843.6			843.6	929.1
Plastic tipped legs	C				393.3			393.3	433.2
Individual high chairs, with plate (HCP), 5" high	C				214.32			214.32	235.98
8" high	C				296.4			296.4	326.04
Bar chair (BC), 1-1/2" high, plain steel	C				45.6			45.6	50.16
Galvanized	C				51.3			51.3	56.43
Stainless tipped legs	C				530.1			530.1	581.4
Plastic tipped legs	C				49.02			49.02	54.15
Continuous high chairs (CHC), legs 8" OC, 4" high, plain steel	C.L.F.				61.56			61.56	67.83
Galvanized	C.L.F.				74.1			74.1	81.51
Stainless tipped legs	C.L.F.				547.2			547.2	598.5
Plastic tipped legs	C.L.F.				82.08			82.08	90.06
Epoxy dipped	C.L.F.				106.02			106.02	116.28
6" high, plain	C.L.F.				84.36			84.36	92.91
Galvanized	C.L.F.				101.46			101.46	111.15
Stainless tipped legs	C.L.F.				570			570	627
Plastic tipped legs	C.L.F.				112.86			112.86	124.26
Epoxy dipped	C.L.F.				144.78			144.78	159.6
8" high, plain	C.L.F.				119.7			119.7	132.24
Galvanized	C.L.F.				143.64			143.64	158.46
Stainless tipped legs	C.L.F.				604.2			604.2	666.9
Plastic tipped legs	C.L.F.				142.5			142.5	156.18
Epoxy dipped	C.L.F.				183.54			183.54	201.78
For continuous bottom wire runners, add	C.L.F.				33.06			33.06	36.48
For continuous bottom plate, add	C.L.F.				226.86			226.86	249.66
Screed chair base, 1/2" coil thread diam., 2-1/2" high, plain steel	C				381.9			381.9	421.8
Galvanized	C				461.7			461.7	507.3
5-1/2" high, plain	C				461.7			461.7	507.3
Galvanized	C				552.9			552.9	609.9
3/4" coil thread diam., 2-1/2" high, plain steel	C				478.8			478.8	530.1
Galvanized	C				575.7			575.7	632.7
5-1/2" high, plain steel	C				592.8			592.8	649.8
Galvanized	C				712.5			712.5	780.9
Screed holder, 1/2" coil thread diam. for pipe screed, plain steel, 6" long	C				461.7			461.7	507.3
12" long	C				712.5			712.5	780.9
3/4" coil thread diam. for pipe screed, plain steel, 6" long	C				666.9			666.9	729.6
12" long	C				1065.9			1065.9	1168.5
Screw anchor for bolts, plain steel, 3/4" diameter x 4" long	C				644.1			644.1	706.8
1" diameter x 6" long	C				1060.2			1060.2	1168.5
1-1/2" diameter x 8" long	C				1339.5			1339.5	1482
Screw anchor eye bolts, 3/4" x 3" long	C				3363			3363	3705
1" x 3-1/2" long	C				4503			4503	4959
1-1/2" x 6" long	C				13794			13794	15162
Screw anchor bolts, 3/4" x 9" long	C				1938			1938	2109
1" x 12" long	C				3705			3705	4075.5
Slab lifting inserts, single pickup, galv, 3/4" diam., 5" high	C				2080.5			2080.5	2280
6" high	C				2109			2109	2308.5
7" high	C				2137.5			2137.5	2365.5
1" diameter, 5-1/2" high	C				2194.5			2194.5	2422.5
7" high	C				2251.5			2251.5	2479.5

Double pickup lifting inserts, 1" diameter, 5-1/2" high	C	4218	4218	4645.5
7" high	C	4674	4674	5158.5
1-1/2" diameter, 8" high	C	5899.5	5899.5	6469.5
Subgrade chairs, #4 bar head, 3-1/2" high	C	45.6	45.6	50.16
12" high	C	53.58	53.58	58.71
#6 bar head, 3-1/2" high	C	45.6	45.6	50.16
12" high	C	53.58	53.58	58.71
Subgrade stakes, no nail holes, 3/4" diameter, 12" long	C	165.3	165.3	182.4
24" long	C	273.6	273.6	300.96
7/8" diameter, 12" long	C	444.6	444.6	490.2
24" long	C	752.4	752.4	826.5
Tie wire, 16 ga. annealed steel	Cwt.	193.8	193.8	213.18

SPlicing REINFORCING BARS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Including holding bars in place while splicing									
Standard, self-aligning type, taper threaded, #4 bars	Ea.	C25	190	0.168	8.27	6.1		14.37	19.64
#5 bars	Ea.	C25	170	0.188	10.32	6.83		17.15	23.09
#6 bars	Ea.	C25	150	0.213	11.63	7.72		19.35	26.15
#7 bars	Ea.	C25	130	0.246	13.45	8.9		22.35	30.15
#8 bars	Ea.	C25	115	0.278	22.29	10.07		32.36	41.91
#9 bars	Ea.	C5	105	0.457	26.22	19.02	5.81	51.05	66.82
#10 bars	Ea.	C5	95	0.505	29.07	21.04	6.43	56.54	74.27
#11 bars	Ea.	C5	85	0.565	30.78	23.5	7.21	61.49	80.73
#14 bars	Ea.	C5	65	0.738	39.9	30.77	9.44	80.11	105.17
#18 bars	Ea.	C5	45	1.067	60.99	44.2	13.58	118.77	156.09
Transition self-aligning, taper threaded, #18-14	Ea.	C5	45	1.067	80.94	44.2	13.58	138.72	177.75
#18-11	Ea.	C5	45	1.067	82.08	44.2	13.58	139.86	179.46
#14-11	Ea.	C5	65	0.738	54.15	30.77	9.44	94.36	121.13
#11-10	Ea.	C5	85	0.565	37.62	23.5	7.21	68.33	88.71
#10-9	Ea.	C5	95	0.505	35.91	21.04	6.43	63.38	81.68
#9-8	Ea.	C25	105	0.305	32.49	11.02		43.51	54.99
#8-7	Ea.	C25	115	0.278	30.21	10.07		40.28	50.46
#7-6	Ea.	C25	130	0.246	20.86	8.9		29.76	38.19
Position coupler for curved bars, taper threaded, #4 bars	Ea.	C25	160	0.2	39.33	7.27		46.6	55.85
#5 bars	Ea.	C25	145	0.221	41.61	8		49.61	59.42
#6 bars	Ea.	C25	130	0.246	50.16	8.9		59.06	70.11
#7 bars	Ea.	C25	110	0.291	53.01	10.52		63.53	76.89
#8 bars	Ea.	C25	100	0.32	55.29	11.58		66.87	80.45
#9 bars	Ea.	C5	90	0.533	60.42	22.21	6.79	89.42	110.52
#10 bars	Ea.	C5	80	0.6	64.98	25.18	7.67	97.83	121.05
#11 bars	Ea.	C5	70	0.686	67.83	28.53	8.76	105.12	131.3
#14 bars	Ea.	C5	55	0.873	84.36	36.37	11.15	131.88	165.58
#18 bars	Ea.	C5	40	1.2	104.31	49.8	15.3	169.41	215.36
Transition position coupler for curved bars, taper threaded, #18-14	Ea.	C5	40	1.2	137.94	49.8	15.3	203.04	251.84
#18-11	Ea.	C5	40	1.2	139.08	49.8	15.3	204.18	254.12
#14-11	Ea.	C5	55	0.873	96.9	36.37	11.15	144.42	179.26
#11-10	Ea.	C5	70	0.686	78.66	28.53	8.76	115.95	143.27
#10-9	Ea.	C5	80	0.6	75.24	25.18	7.67	108.09	132.45
#9-8	Ea.	C25	90	0.356	70.11	12.87		82.98	99.73
#8-7	Ea.	C25	100	0.32	64.41	11.58		75.99	90.71
#7-6	Ea.	C25	110	0.291	62.7	10.52		73.22	87.15
Sleeve type w/grout filler, for precast concrete, #6 bars	Ea.	C25	72	0.444	33.06	16.11		49.17	63.89
#7 bars	Ea.	C25	64	0.5	39.33	18.13		57.46	74.08
#8 bars	Ea.	C25	56	0.571	46.74	20.7		67.44	87.11
#9 bars	Ea.	C25	48	0.667	53.58	24.06		77.64	101.24
#10 bars	Ea.	C5	40	1.2	65.55	49.8	15.3	130.65	172.04
#11 bars	Ea.	C5	32	1.5	72.39	62.66	19.13	154.18	205.13
#14 bars	Ea.	C5	24	2	112.29	83.37	25.41	221.07	291.02
Sleeve type w/ferrous filler, for critical structures, #6 bars	Ea.	C25	72	0.444	103.74	16.11		119.85	141.98
#7 bars	Ea.	C25	64	0.5	105.45	18.13		123.58	147.61
#8 bars	Ea.	C25	56	0.571	110.58	20.7		131.28	157.79
#9 bars	Ea.	C5	48	1	115.14	41.4	12.76	169.3	209.97
#10 bars	Ea.	C5	40	1.2	120.84	49.8	15.3	185.94	233.6
#11 bars	Ea.	C5	32	1.5	147.06	62.66	19.13	228.85	286.07
#14 bars	Ea.	C5	24	2	183.54	83.37	25.41	292.32	368.54
#18 bars	Ea.	C5	16	3	265.62	125.33	38.37	429.32	543.11
Weldable half coupler, taper threaded, #4 bars	Ea.	E16	120	0.133	12.94	6.21	0.85	20	26.09
#5 bars	Ea.	E16	112	0.143	15.22	6.6	0.91	22.73	29.46
#6 bars	Ea.	E16	104	0.154	23.94	7.16	0.99	32.09	40.51
#7 bars	Ea.	E16	96	0.167	27.93	7.72	1.07	36.72	45.66
#8 bars	Ea.	E16	88	0.182	29.07	8.45	1.16	38.68	48.76
#9 bars	Ea.	E16	80	0.2	32.49	9.29	1.28	43.06	53.19
#10 bars	Ea.	E16	72	0.222	33.06	10.29	1.42	44.77	56.34
#11 bars	Ea.	E16	64	0.25	35.34	11.58	1.6	48.52	61.1
#14 bars	Ea.	E16	56	0.286	40.47	13.26	1.83	55.56	70.53
#18 bars	Ea.	E16	48	0.333	66.12	15.44	2.13	83.69	102.72

Plain Steel Reinforcement Bars

REINFORCING STEEL, MILL BASE PLUS EXTRAS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Reinforcing, A615 grade 40, mill base	Ton				763.8			763.8	843.6
Detailed, cut, bent, and delivered	Ton				1094.4			1094.4	1197
Reinforcing steel, A615 grade 60, mill base	Ton				763.8			763.8	843.6
Detailed, cut, bent, and delivered	Ton				1094.4			1094.4	1197
Reinforcing steel, extras, included in delivered price									
Mill extra, added for delivery to shop	Ton				40.47			40.47	44.46
Shop extra, added for handling & storage	Ton				51.3			51.3	56.43
Shop extra, added for bending, limited percent of bars	Ton				38.76			38.76	42.75
Average percent of bars	Ton				78.09			78.09	85.5
Large percent of bars	Ton				156.18			156.18	171
Shop extra, added for detailing, under 50 tons	Ton				60.42			60.42	66.12
50 to 150 tons	Ton				45.03			45.03	49.59
150 to 500 tons	Ton				43.32			43.32	47.31
Over 500 tons	Ton				41.04			41.04	45.03
Shop extra, added for listing	Ton				5.81			5.81	6.38
Mill extra, added for quantity, under 20 tons	Ton				25.65			25.65	27.93
Shop extra, added for quantity, under 20 tons	Ton				37.05			37.05	41.04
20 to 49 tons	Ton				27.93			27.93	30.78
50 to 99 tons	Ton				18.58			18.58	20.41
100 to 300 tons	Ton				11.12			11.12	12.26
Shop extra, added for size, #3	Ton				181.26			181.26	199.5
#4	Ton				90.63			90.63	99.75

#5	Ton				45.03			45.03	49.59
#6	Ton				41.04			41.04	45.03
#7 to #11	Ton				54.15			54.15	59.85
#14	Ton				67.83			67.83	74.67
#18	Ton				76.95			76.95	84.93
Shop extra, added for delivery to job	Ton				20.29			20.29	22.34

REINFORCING IN PLACE, 50-60 ton lots, A615 Grade 60

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes labor, but not material cost, to install accessories									
Made from recycled materials									
Beams & Girders, #3 to #7	Ton	4 Rodm	1.6	20	1094.4	906.39		2000.79	2707.65
#8 to #18	Ton	4 Rodm	2.7	11.852	1094.4	537.12		1631.52	2092.2
Columns, #3 to #7	Ton	4 Rodm	1.5	21.333	1094.4	967.94		2062.34	2819.55
#3 to #7, alternate method	Lb.	4 Rodm	3000	0.011	0.57	0.48		1.05	1.44
#8 to #18	Ton	4 Rodm	2.3	13.913	1094.4	632.24		1726.64	2248.86
#8 to #18, alternate method	Lb.	4 Rodm	4600	0.007	0.57	0.31		0.88	1.16
Spirals, hot rolled, 8" to 15" diameter	Ton	4 Rodm	2.2	14.545	1795.5	660.21		2455.71	3063.12
15" to 24" diameter	Ton	4 Rodm	2.2	14.545	1710	660.21		2370.21	2977.62
24" to 36" diameter	Ton	4 Rodm	2.3	13.913	1624.5	632.24		2256.74	2847.36
36" to 48" diameter	Ton	4 Rodm	2.4	13.333	1539	604.26		2143.26	2717.1
48" to 64" diameter	Ton	4 Rodm	2.5	12.8	1710	581.88		2291.88	2848.94
64" to 84" diameter	Ton	4 Rodm	2.6	12.308	1795.5	559.5		2355	2895.27
84" to 96" diameter	Ton	4 Rodm	2.7	11.852	1881	537.12		2418.12	2947.2
Elevated slabs, #4 to #7	Ton	4 Rodm	2.9	11.034	1094.4	497.96		1592.36	2030.66
Footings, #4 to #7	Ton	4 Rodm	2.1	15.238	1094.4	688.19		1782.59	2343.98
Footings, #4 to #7	Lb.	4 Rodm	4200	0.008	0.55	0.35		0.9	1.17
#8 to #18	Ton	4 Rodm	3.6	8.889	1094.4	402.84		1497.24	1868.4
#8 to #18	Lb.	4 Rodm	7200	0.004	0.55	0.2		0.75	0.94
Slab on grade, #3 to #7	Ton	4 Rodm	2.3	13.913	1094.4	632.24		1726.64	2248.86
Slab on grade, #3 to #7	Lb.	4 Rodm	4200	0.008	0.55	0.35		0.9	1.17
Walls, #3 to #7	Ton	4 Rodm	3	10.667	1094.4	481.17		1575.57	2002.68
#8 to #18	Ton	4 Rodm	4	8	1094.4	363.68		1458.08	1801.26

For other than 50-60 ton lots

Under 10 ton job, #3 to #7, add					25.00%	10.00%			
#8 to #18, add					20.00%	10.00%			
10-50 ton job, #3 to #7, add					10.00%				
#8 to #18, add					5.00%				
60-100 ton job, #3 to #7, deduct					5.00%				
#8 to #18, deduct					10.00%				
Over 100 ton job, #3 to #7, deduct					10.00%				
#8 to #18, deduct					15.00%				
Reinforcing in place, A615 Grade 75, add	Ton				104.31			104.31	115.14
Grade 90, add	Ton				139.08			139.08	152.76
Unloading & sorting, add to above	Ton	C5	100	0.48		19.97	6.12	26.09	39.75
Crane cost for handling, 90 picks/day, up to 1.5 tons/bundle, add to above	Ton	C5	135	0.356		14.83	4.53	19.36	29.61
1.0 ton/bundle	Ton	C5	92	0.522		21.71	6.64	28.35	43.68
0.5 ton/bundle	Ton	C5	35	1.371		57.07	17.47	74.54	114.36
Dowels, 2 feet long, deformed, #3	Ea.	2 Rodm	520	0.031	0.46	1.39		1.85	2.83
#4	Ea.	2 Rodm	480	0.033	0.8	1.51		2.31	3.4
#5	Ea.	2 Rodm	435	0.037	1.25	1.67		2.92	4.16
#6	Ea.	2 Rodm	360	0.044	1.8	2.01		3.81	5.34
Longer and heavier dowels, add	Lb.	2 Rodm	725	0.022	0.6	1		1.6	2.33
Smooth dowels, 12" long, 1/4" or 3/8" diameter	Ea.	2 Rodm	140	0.114	0.83	5.17		6	9.53
5/8" diameter	Ea.	2 Rodm	125	0.128	1.46	5.82		7.28	11.28
3/4" diameter	Ea.	2 Rodm	110	0.145	1.8	6.6		8.4	12.95
Dowel sleeves for CIP concrete, 2-part system									
Sleeve base, plastic, for 5/8" smooth dowel sleeve, fasten to edge form	Ea.	1 Rodm	200	0.04	0.6	1.81		2.41	3.68
Sleeve, plastic, 12" long, for 5/8" smooth dowel, snap onto base	Ea.	1 Rodm	400	0.02	1.52	0.91		2.43	3.17
Sleeve base, for 3/4" smooth dowel sleeve	Ea.	1 Rodm	175	0.046	0.6	2.07		2.67	4.11
Sleeve, 12" long, for 3/4" smooth dowel	Ea.	1 Rodm	350	0.023	1.38	1.03		2.41	3.24
Sleeve base, for 1" smooth dowel sleeve	Ea.	1 Rodm	150	0.053	0.78	2.42		3.2	4.89
Sleeve, 12" long, for 1" smooth dowel	Ea.	1 Rodm	300	0.027	1.62	1.21		2.83	3.79
Dowel caps, visual warning only, plastic, #3 to #8	Ea.	2 Rodm	800	0.02	0.34	0.91		1.25	1.89
#8 to #18	Ea.	2 Rodm	750	0.021	0.84	0.96		1.8	2.54
Impalement protective, plastic, #4 to #9	Ea.	2 Rodm	800	0.02	1.23	0.91		2.14	2.87

Galvanized Reinforcement Steel Bars

GALVANIZED REINFORCING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Add to plain steel rebar pricing for galvanized rebar	Ton				524.4			524.4	575.7

Epoxy-Coated Reinforcement Steel Bars

EPOXY-COATED REINFORCING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Add to plain steel rebar pricing for epoxy-coated rebar	Ton				894.9			894.9	986.1

Stainless Steel Reinforcement Bars

STAINLESS STEEL REINFORCING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Add to plain steel rebar pricing for stainless steel rebar					300.00%				

Composite Reinforcement Bars

GLASS FIBER-REINFORCED POLYMER REINFORCEMENT BARS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes labor, but not material cost, to install accessories									
#2 bar, .043 lb./L.F.	L.F.	4 Rodm	9500	0.003	0.46	0.16		0.62	0.76
#3 bar, .092 lb./L.F.	L.F.	4 Rodm	9300	0.003	0.72	0.16		0.88	1.05
#4 bar, .160 lb./L.F.	L.F.	4 Rodm	9100	0.004	1.04	0.16		1.2	1.41
#5 bar, .258 lb./L.F.	L.F.	4 Rodm	8700	0.004	1.24	0.17		1.41	1.65
#6 bar, .372 lb./L.F.	L.F.	4 Rodm	8300	0.004	2.06	0.18		2.24	2.56
#7 bar, .497 lb./L.F.	L.F.	4 Rodm	7900	0.004	2.7	0.18		2.88	3.28
#8 bar, .620 lb./L.F.	L.F.	4 Rodm	7400	0.004	2.71	0.19		2.9	3.31
#9 bar, .800 lb./L.F.	L.F.	4 Rodm	6800	0.005	4.55	0.21		4.76	5.36
#10 bar, 1.08 lb./L.F.	L.F.	4 Rodm	5800	0.006	4.28	0.25		4.53	5.12
For bends, add per bend	Ea.				1.76			1.76	1.93

Fabric and Grid Reinforcing

Plain Welded Wire Fabric Reinforcing

PLAIN WELDED WIRE FABRIC ASTM A185

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes labor, but not material cost, to install accessories									
Made from recycled materials									
Sheets									
6 x 6 - W1.4 x W1.4 (10 x 10) 21 lb./C.S.F.	C.S.F.	2 Rodm	35	0.457	17.21	20.7		37.91	53.61
6 x 6 - W2.1 x W2.1 (8 x 8) 30 lb./C.S.F.	C.S.F.	2 Rodm	31	0.516	22.34	23.5		45.84	63.68
6 x 6 - W2.9 x W2.9 (6 x 6) 42 lb./C.S.F.	C.S.F.	2 Rodm	29	0.552	29.07	25.18		54.25	73.32

Beams (3500 psi), 5 kip/L.F., 10' span	C.Y.	C14A	15.62	12.804	348.5	527.68	60.15	936.33	1321.46
25' span	C.Y.	C14A	18.55	10.782	363.88	440.64	50.81	855.33	1190.15
Chimney foundations (5000 psi), over 5 C.Y.	C.Y.	C14C	32.22	3.476	175.28	136	0.83	312.11	418.82
(3500 psi), under 5 C.Y.	C.Y.	C14C	23.71	4.724	202.95	184.96	1.13	389.04	530.47
Columns, square (4000 psi), 12" x 12", up to 1% reinforcing by area	C.Y.	C14A	11.96	16.722	389.5	685.44	78.81	1153.75	1654.37
Up to 2% reinforcing by area	C.Y.	C14A	10.13	19.743	594.5	810.56	92.81	1497.87	2113.02
Up to 3% reinforcing by area	C.Y.	C14A	9.03	22.148	871.25	908.48	104.74	1884.47	2596.69
16" x 16", up to 1% reinforcing by area	C.Y.	C14A	16.22	12.33	312.63	505.92	58.07	876.62	1250.36
Up to 2% reinforcing by area	C.Y.	C14A	12.57	15.911	507.38	652.8	74.66	1234.84	1729.07
Up to 3% reinforcing by area	C.Y.	C14A	10.25	19.512	768.75	799.68	91.77	1660.2	2279.54
24" x 24", up to 1% reinforcing by area	C.Y.	C14A	23.66	8.453	267.53	348.16	39.92	655.61	914.37
Up to 2% reinforcing by area	C.Y.	C14A	17.71	11.293	451	462.4	53.41	966.81	1328.2
Up to 3% reinforcing by area	C.Y.	C14A	14.15	14.134	702.13	582.08	66.37	1350.58	1809.87
36" x 36", up to 1% reinforcing by area	C.Y.	C14A	33.69	5.936	237.8	243.71	28	509.51	699.97
Up to 2% reinforcing by area	C.Y.	C14A	23.32	8.576	399.75	353.6	40.44	793.79	1072.86
Up to 3% reinforcing by area	C.Y.	C14A	17.82	11.223	656	462.4	52.89	1171.29	1547.74
Columns, round (4000 psi), tied, 12" diameter, up to 1% reinforcing by area	C.Y.	C14A	20.97	9.537	389.5	391.68	45.11	826.29	1127.44
Up to 2% reinforcing by area	C.Y.	C14A	15.27	13.098	589.38	538.56	61.7	1189.64	1610.96
Up to 3% reinforcing by area	C.Y.	C14A	12.11	16.515	855.88	680	77.78	1613.66	2143.75
16" diameter, up to 1% reinforcing by area	C.Y.	C14A	31.49	6.351	328	261.12	30.07	619.19	826.62
Up to 2% reinforcing by area	C.Y.	C14A	19.12	10.46	533	429.76	49.26	1012.02	1350.81
Up to 3% reinforcing by area	C.Y.	C14A	13.77	14.524	773.88	598.4	68.44	1440.72	1921.14
20" diameter, up to 1% reinforcing by area	C.Y.	C14A	41.04	4.873	317.75	200.19	22.81	540.75	705.75
Up to 2% reinforcing by area	C.Y.	C14A	24.05	8.316	507.38	342.72	39.41	889.51	1162.3
Up to 3% reinforcing by area	C.Y.	C14A	17.01	11.758	768.75	484.16	55.48	1308.39	1711.41
24" diameter, up to 1% reinforcing by area	C.Y.	C14A	51.85	3.857	297.25	158.85	18.15	474.25	611.26
Up to 2% reinforcing by area	C.Y.	C14A	27.06	7.391	502.25	303.55	34.74	840.54	1097.79
Up to 3% reinforcing by area	C.Y.	C14A	18.29	10.935	748.25	451.52	51.33	1251.1	1626.93
36" diameter, up to 1% reinforcing by area	C.Y.	C14A	75.04	2.665	303.4	109.89	12.55	425.84	528.62
Up to 2% reinforcing by area	C.Y.	C14A	37.49	5.335	486.88	219.78	24.89	731.55	924.96
Up to 3% reinforcing by area	C.Y.	C14A	22.84	8.757	738	359.04	41.48	1138.52	1453.26
Elevated slab (4000 psi), flat slab with drops, 125 psf Sup. Load, 20' span	C.Y.	C14B	38.45	5.41	291.1	221.95	24.37	537.42	714.63
30' span	C.Y.	C14B	50.99	4.079	305.45	167.55	18.46	491.46	630.9
Flat plate, 125 psf Sup. Load, 15' span	C.Y.	C14B	30.24	6.878	267.53	282.88	31.11	581.52	796.24
25' span	C.Y.	C14B	49.6	4.194	276.75	171.9	18.98	467.63	611.31
Waffle const., 30" domes, 125 psf Sup. Load, 20' span	C.Y.	C14B	37.07	5.611	304.43	230.66	25.41	560.5	741.93
30' span	C.Y.	C14B	44.07	4.72	281.88	193.66	21.26	496.8	658.01
One way joists, 30" pans, 125 psf Sup. Load, 15' span	C.Y.	C14B	27.38	7.597	410	312.26	34.22	756.48	1005.65
25' span	C.Y.	C14B	31.15	6.677	374.13	274.18	30.07	678.38	905.27
One way beam & slab, 125 psf Sup. Load, 15' span	C.Y.	C14B	20.59	10.102	285.98	413.44	45.63	745.05	1053.8
25' span	C.Y.	C14B	28.36	7.334	271.63	301.38	33.18	606.19	835.06
Two way beam & slab, 125 psf Sup. Load, 15' span	C.Y.	C14B	24.04	8.652	276.75	353.6	38.89	669.24	934.99
25' span	C.Y.	C14B	35.87	5.799	241.9	238.27	26.44	506.61	691.64
Elevated slabs, flat plate, including finish, not including forms or reinforcing									
Regular concrete (4000 psi), 4" slab	S.F.	C8	2613	0.021	1.74	0.77	0.35	2.86	3.57
6" slab	S.F.	C8	2585	0.022	2.54	0.78	0.35	3.67	4.47
2-1/2" thick floor fill	S.F.	C8	2685	0.021	1.14	0.75	0.34	2.23	2.86
Lightweight, 110 #/C.F., 2-1/2" thick floor fill	S.F.	C8	2585	0.022	1.17	0.78	0.35	2.3	2.96
Cellular concrete, 1-5/8" fill, under 5000 S.F.	S.F.	C8	2000	0.028	0.81	1.01	0.46	2.28	3.05
Over 10,000 S.F.	S.F.	C8	2200	0.025	0.77	0.91	0.41	2.09	2.82
Add per floor for 3 to 6 stories high	S.F.	C8	31800	0.002		0.07	0.03	0.1	0.14
For 7 to 20 stories high	S.F.	C8	21200	0.003		0.1	0.04	0.14	0.2
Equipment pad (3000 psi), 3' x 3' x 6" thick	Ea.	C14H	45	1.067	45.61	42.98	0.59	89.18	122.14
4' x 4' x 6" thick	Ea.	C14H	30	1.6	70.73	64.19	0.88	135.8	185.49
5' x 5' x 8" thick	Ea.	C14H	18	2.667	129.15	107.17	1.46	237.78	322.54
6' x 6' x 8" thick	Ea.	C14H	14	3.429	177.33	138.18	1.89	317.4	425.32
8' x 8' x 10" thick	Ea.	C14H	8	6	379.25	241.54	3.3	624.09	821.36
10' x 10' x 12" thick	Ea.	C14H	5	9.6	661.13	386.24	5.29	1052.66	1375.53
Flexural concrete on grade, direct chute, 500 psi, no forms, reinf, finish	C.Y.	C8A	150	0.32	124.03	11.15		135.18	154.72
650 psi	C.Y.	C8A	150	0.32	137.35	11.15		148.5	170.09
750 psi	C.Y.	C8A	150	0.32	207.05	11.15		218.2	245.94
Pumped, 500 psi	C.Y.	C8	70	0.8	124.03	28.83	13.07	165.93	198.02
650 psi	C.Y.	C8	70	0.8	137.35	28.83	13.07	179.25	213.39
750 psi	C.Y.	C8	70	0.8	207.05	28.83	13.07	248.95	289.24
Footings (3000 psi), spread under 1 C.Y.	C.Y.	C14C	28	4	189.63	156.67	0.95	347.25	470.17
1 C.Y. to 5 C.Y.	C.Y.	C14C	43	2.605	224.48	101.73	0.62	326.83	416.34
Over 5 C.Y.	C.Y.	C14C	75	1.493	208.08	58.21	0.35	266.64	325.79
Footings, strip (3000 psi), 18" x 9", unreinforced	C.Y.	C14L	40	2.4	149.65	91.94	0.67	242.26	318.08
18" x 9", reinforced	C.Y.	C14C	35	3.2	173.23	125.12	0.77	299.12	399.29
20" x 10", unreinforced	C.Y.	C14L	45	2.133	145.55	81.6	0.6	227.75	297.57
20" x 10", reinforced	C.Y.	C14C	40	2.8	165.03	109.89	0.66	275.58	363.86
24" x 12", unreinforced	C.Y.	C14L	55	1.745	143.5	66.91	0.49	210.9	269.36
24" x 12", reinforced	C.Y.	C14C	48	2.333	162.98	91.39	0.56	254.93	331.21
36" x 12", unreinforced	C.Y.	C14L	70	1.371	139.4	52.77	0.38	192.55	241.2
36" x 12", reinforced	C.Y.	C14C	60	1.867	156.83	72.9	0.45	230.18	293.46
Foundation mat (3000 psi), under 10 C.Y.	C.Y.	C14C	38.67	2.896	226.53	113.15	0.69	340.37	438.05
Over 20 C.Y.	C.Y.	C14C	56.4	1.986	200.9	77.79	0.48	279.17	351.39
Wall, free-standing (3000 psi), 8" thick, 8' high	C.Y.	C14D	45.83	4.364	186.55	178.43	20.53	385.51	524.83
14' high	C.Y.	C14D	27.26	7.337	216.28	300.29	34.74	551.31	776.13
12" thick, 8' high	C.Y.	C14D	64.32	3.109	169.13	127.3	14.62	311.05	413.69
14' high	C.Y.	C14D	40.01	4.999	179.38	204.54	23.33	407.25	560.01
15" thick, 8' high	C.Y.	C14D	80.02	2.499	164	102.27	11.77	278.04	362.02
12' high	C.Y.	C14D	51.26	3.902	164	159.94	18.35	342.29	465.02
18' high	C.Y.	C14D	48.85	4.094	182.45	167.55	19.29	369.29	500.69
Handicap access ramp (4000 psi), railing both sides, 3' wide	L.F.	C14H	14.58	3.292	369	132.74	1.8	503.54	626.67
5' wide	L.F.	C14H	12.22	3.928	379.25	157.76	2.16	539.17	684.87
With 6" curb and rails both sides, 3' wide	L.F.	C14H	8.55	5.614	379.25	226.3	3.09	608.64	799.04
5' wide	L.F.	C14H	7.31	6.566	384.38	264.38	3.61	652.37	870.03
Slab on grade (3500 psi), not including finish, 4" thick	C.Y.	C14E	60.75	1.449	148.63	57.66	0.44	206.73	260.22
6" thick	C.Y.	C14E	92	0.957	143.5	38.08	0.29	181.87	221.27
Thickened slab edge (3500 psi), for slab on grade poured monolithically with slab; depth is in addition to slab thickness; formed vertical outside edge, earthen bottom and inside slope									
8" deep x 8" wide bottom, unreinforced	L.F.	C14L	2190	0.044	4.07	1.68	0.01	5.76	7.27
8" x 8", reinforced	L.F.	C14C	1670	0.067	6.36	2.62	0.02	9	11.39
12" deep x 12" wide bottom, unreinforced	L.F.	C14L	1800	0.053	8.35	2.05	0.01	10.41	12.57
12" x 12", reinforced	L.F.	C14C	1310	0.086	12.56	3.34	0.02	15.92	19.41
16" deep x 16" wide bottom, unreinforced	L.F.	C14L	1440	0.067	14.15	2.56	0.02	16.73	19.83
16" x 16", reinforced	L.F.	C14C	1120	0.1	19.17	3.91	0.02	23.1	27.51
20" deep x 20" wide bottom, unreinforced	L.F.	C14L	1150	0.083	21.53	3.2	0.02	24.75	28.9

20" x 20", reinforced	L.F.	C14C	920	0.122	27.68	4.75	0.03	32.46	38.67
24" deep x 24" wide bottom, unreinforced	L.F.	C14L	930	0.103	30.24	3.95	0.03	34.22	39.92
24" x 24", reinforced	L.F.	C14C	740	0.151	38.95	5.93	0.03	44.91	52.43
Slab on grade (3500 psi), incl. troweled finish, not incl. forms									
or reinforcing, over 10,000 S.F., 4" thick	S.F.	C14F	3425	0.021	1.65	0.78	0.01	2.44	3.1
6" thick	S.F.	C14F	3350	0.021	2.42	0.79	0.01	3.22	3.99
8" thick	S.F.	C14F	3184	0.023	3.31	0.84	0.01	4.16	5.02
12" thick	S.F.	C14F	2734	0.026	4.96	0.98	0.01	5.95	7.09
15" thick	S.F.	C14F	2505	0.029	6.25	1.07	0.01	7.33	8.62
Slab on grade (3000 psi), incl. broom finish, not incl. forms									
or reinforcing, 4" thick	S.F.	C14G	2873	0.019	1.61	0.71	0.01	2.33	2.94
6" thick	S.F.	C14G	2590	0.022	2.52	0.79	0.01	3.32	4.07
8" thick	S.F.	C14G	2320	0.024	3.28	0.88	0.01	4.17	5.07
Lift slab in place above the foundation, incl. forms, reinforcing,									
concrete (4000 psi) and columns, over 20,000 S.F./floor	S.F.	C14B	2113	0.098	7.64	4.04	0.45	12.13	15.59
10,000 S.F. to 20,000 S.F./floor	S.F.	C14B	1650	0.126	8.3	5.18	0.57	14.05	18.35
Under 10,000 S.F./floor	S.F.	C14B	1500	0.139	8.97	5.71	0.62	15.3	20.05
Lightweight, ready mix, including screed finish only, not including forms or reinforcing									
1:4 (2500 psi) for structural roof decks	C.Y.	C14B	260	0.8	118.9	32.64	3.62	155.16	188.56
1:6 (3000 psi) for ground slab with radiant heat	C.Y.	C14F	92	0.783	130.18	28.83	0.29	159.3	190.13
1:3:2 (2000 psi) with sand aggregate, roof deck	C.Y.	C14B	260	0.8	117.88	32.64	3.62	154.14	187.53
Ground slab (2000 psi)	C.Y.	C14F	107	0.673	117.88	25.02	0.25	143.15	170.22
Pile caps (3000 psi), incl. forms and reinf., sq. or rect., under 10 C.Y.	C.Y.	C14C	54.14	2.069	189.63	81.06	0.5	271.19	343.46
Over 10 C.Y.	C.Y.	C14C	75	1.493	179.38	58.21	0.35	237.94	295.04
Triangular or hexagonal, under 10 C.Y.	C.Y.	C14C	53	2.113	145.55	82.69	0.51	228.75	298.57
Over 10 C.Y.	C.Y.	C14C	85	1.318	161.95	51.68	0.31	213.94	263.08
Retaining walls (3000 psi), gravity, 4' high see Section 32 32	C.Y.	C14D	66.2	3.021	164	124.03	14.21	302.24	401.69
10' high	C.Y.	C14D	125	1.6	157.85	65.28	7.52	230.65	290.33
Cantilever, level backfill loading, 8' high	C.Y.	C14D	70	2.857	175.28	116.42	13.43	305.13	401.14
16' high	C.Y.	C14D	91	2.198	169.13	89.76	10.37	269.26	346.97
Stairs (3500 psi), not including safety treads, free standing, 3'-6" wide	LF Nose	C14H	83	0.578	6.2	23.39	0.32	29.91	45.79
Cast on ground	LF Nose	C14H	125	0.384	5.28	15.45	0.21	20.94	31.65
Stair landings, free standing	S.F.	C14H	200	0.24	4.95	9.68	0.13	14.76	21.57
Cast on ground	S.F.	C14H	475	0.101	4.03	4.07	0.05	8.15	11.25

Structural Concrete

Heavyweight Structural Concrete

CONCRETE, HAND MIX for small quantities or remote areas

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes bulk local aggregate, bulk sand, bagged Portland cement (Type I) and water, using gas powered cement mixer									
2500 psi	C.F.	C30	135	0.059	3.9	1.9	1.23	7.03	8.81
3000 psi	C.F.	C30	135	0.059	4.21	1.9	1.23	7.34	9.16
3500 psi	C.F.	C30	135	0.059	4.4	1.9	1.23	7.53	9.37
4000 psi	C.F.	C30	135	0.059	4.61	1.9	1.23	7.74	9.6
4500 psi	C.F.	C30	135	0.059	4.87	1.9	1.23	8	9.86
5000 psi	C.F.	C30	135	0.059	5.18	1.9	1.23	8.31	10.27
Using pre-bagged dry mix and wheelbarrow (80-lb. bag = 0.6 C.F.)									
4000 psi	C.F.	1 Clab	48	0.167	7.02	5.35		12.37	16.66

CONCRETE, VOLUMETRIC SITE-MIXED

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Mixed on-site in volumetric truck									
Includes local aggregate, sand, Portland cement (Type I) and water									
Excludes all additives and treatments									
3000 psi, 1 C.Y. mixed and discharged	C.Y.				219.35			219.35	240.88
2 C.Y.	C.Y.				161.95			161.95	178.35
3 C.Y.	C.Y.				141.45			141.45	155.8
4 C.Y.	C.Y.				128.13			128.13	140.43
5 C.Y.	C.Y.				114.8			114.8	126.08
For truck holding/waiting time past first 2 on-site hours, add	Hr.				95.84			95.84	105.58
For trip charge beyond first 20 miles, each way, add	Mile				3.73			3.73	4.1
For each additional increase of 500 psi, add	Ea.				4.27			4.27	4.7

HEAVYWEIGHT CONCRETE, READY MIX, delivered

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes local aggregate, sand, Portland cement (Type I) and water									
Excludes all additives and treatments									
2000 psi	C.Y.				117.88			117.88	130.18
2500 psi	C.Y.				121.98			121.98	133.25
3000 psi	C.Y.				124.03			124.03	136.33
3500 psi	C.Y.				127.1			127.1	140.43
4000 psi	C.Y.				131.2			131.2	143.5
4500 psi	C.Y.				134.28			134.28	147.6
5000 psi	C.Y.				137.35			137.35	151.7
6000 psi	C.Y.				141.45			141.45	155.8
8000 psi	C.Y.				148.63			148.63	164
10,000 psi	C.Y.				156.83			156.83	172.2
12,000 psi	C.Y.				164			164	180.4
1:4 topping	C.Y.				110.7			110.7	121.98
1:3 topping	C.Y.				119.93			119.93	132.23
1:2 topping	C.Y.				134.28			134.28	147.6
High early strength, 2000 psi	C.Y.				125.05			125.05	137.35
2500 psi	C.Y.				129.15			129.15	141.45
3000 psi	C.Y.				133.25			133.25	147.6
3500 psi	C.Y.				137.35			137.35	151.7
4000 psi	C.Y.				141.45			141.45	155.8
4500 psi	C.Y.				144.53			144.53	159.9
5000 psi	C.Y.				148.63			148.63	164
6000 psi	C.Y.				153.75			153.75	170.15
8000 psi	C.Y.				165.03			165.03	181.43
10,000 psi	C.Y.				176.3			176.3	193.73
12,000 psi	C.Y.				187.58			187.58	206.03
For high early strength (Portland cement Type III), add	C.Y.				10.00%				
For structural lightweight with regular sand, add	C.Y.				25.00%				
For winter concrete (hot water), add	C.Y.				5.48			5.48	6.05
For mid-range water reducer, add	C.Y.				3.68			3.68	4.05
For high-range water reducer/superplasticizer, add	C.Y.				6.46			6.46	7.07
For retarder, add	C.Y.				3.31			3.31	3.64
For non-Chloride accelerator, add	C.Y.				6.56			6.56	7.23
For Chloride accelerator, per 1%, add	C.Y.				3.96			3.96	4.36
For fiber reinforcing, synthetic (1 lb./C.Y.), add	C.Y.				8.2			8.2	9.02
For Saturday delivery, add	C.Y.				8.71			8.71	9.58

For truck holding/waiting time past 1st hour per load, add	Hr.		107.63		107.63	118.9
For short load (less than 4 C.Y.), add per load	Ea.		88.15		88.15	96.86
For all lightweight aggregate, add	C.Y.		45.00%			
Flowable fill: ash, cement, aggregate, water						
40 - 80 psi	C.Y.		82.51		82.51	91.23
Structural: ash, cement, aggregate, water & sand						
50 psi	C.Y.		82.51		82.51	91.23
140 psi	C.Y.		83.54		83.54	91.74
500 psi	C.Y.		86.1		86.1	94.81
1000 psi	C.Y.		89.18		89.18	98.4

PLACING CONCRETE

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes labor and equipment to place, level (strike off) and consolidate									
Beams, elevated, small beams, pumped	C.Y.	C20	60	1.067		36.99	16.13	53.12	79.2
With crane and bucket	C.Y.	C7	45	1.6		56.03	24.89	80.92	119.98
Large beams, pumped	C.Y.	C20	90	0.711		24.48	10.73	35.21	52.62
With crane and bucket	C.Y.	C7	65	1.108		39.17	17.06	56.23	82.96
Columns, square or round, 12" thick, pumped	C.Y.	C20	60	1.067		36.99	16.13	53.12	79.2
With crane and bucket	C.Y.	C7	40	1.8		63.1	28	91.1	135.04
18" thick, pumped	C.Y.	C20	90	0.711		24.48	10.73	35.21	52.62
With crane and bucket	C.Y.	C7	55	1.309		45.7	20.17	65.87	98.46
24" thick, pumped	C.Y.	C20	92	0.696		23.94	10.53	34.47	51.27
With crane and bucket	C.Y.	C7	70	1.029		35.9	15.87	51.77	77.26
36" thick, pumped	C.Y.	C20	140	0.457		15.83	6.9	22.73	33.73
With crane and bucket	C.Y.	C7	100	0.72		25.02	11.1	36.12	54.07
Elevated slabs, less than 6" thick, pumped	C.Y.	C20	140	0.457		15.83	6.9	22.73	33.73
With crane and bucket	C.Y.	C7	95	0.758		26.66	11.67	38.33	56.92
6" to 10" thick, pumped	C.Y.	C20	160	0.4		13.87	6.07	19.94	29.49
With crane and bucket	C.Y.	C7	110	0.655		22.85	10.11	32.96	49.18
Slabs over 10" thick, pumped	C.Y.	C20	180	0.356		12.35	5.39	17.74	26.31
With crane and bucket	C.Y.	C7	130	0.554		19.42	8.56	27.98	41.48
Footings, continuous, shallow, direct chute	C.Y.	C6	120	0.4		13.49	0.45	13.94	22.79
Pumped	C.Y.	C20	150	0.427		14.8	6.43	21.23	31.58
With crane and bucket	C.Y.	C7	90	0.8		28.29	12.34	40.63	59.77
Footings, continuous, deep, direct chute	C.Y.	C6	140	0.343		11.53	0.38	11.91	19.5
Pumped	C.Y.	C20	160	0.4		13.87	6.07	19.94	29.49
With crane and bucket	C.Y.	C7	110	0.655		22.85	10.11	32.96	49.18
Footings, spread, under 1 C.Y., direct chute	C.Y.	C6	55	0.873		29.38	0.96	30.34	49.48
Pumped	C.Y.	C20	65	0.985		34.27	14.88	49.15	72.96
With crane and bucket	C.Y.	C7	45	1.6		56.03	24.89	80.92	119.98
Over 5 C.Y., direct chute	C.Y.	C6	120	0.4		13.49	0.45	13.94	22.79
Pumped	C.Y.	C20	150	0.427		14.8	6.43	21.23	31.58
With crane and bucket	C.Y.	C7	100	0.72		25.02	11.1	36.12	54.07
Foundation mats, over 20 C.Y., direct chute	C.Y.	C6	350	0.137		4.61	0.16	4.77	7.79
Pumped	C.Y.	C20	400	0.16		5.55	2.42	7.97	11.84
With crane and bucket	C.Y.	C7	300	0.24		8.43	3.7	12.13	18
Grade beams, direct chute	C.Y.	C6	150	0.32		10.77	0.35	11.12	18.22
Pumped	C.Y.	C20	180	0.356		12.35	5.39	17.74	26.31
With crane and bucket	C.Y.	C7	120	0.6		21.05	9.23	30.28	44.98
High rise, for more than 5 stories, pumped, add per story	C.Y.	C20	2100	0.03		1.06	0.46	1.52	2.26
With crane and bucket, add per story	C.Y.	C7	2100	0.034		1.21	0.53	1.74	2.57
Pile caps, under 5 C.Y., direct chute	C.Y.	C6	90	0.533		17.95	0.59	18.54	30.56
Pumped	C.Y.	C20	110	0.582		20.18	8.81	28.99	42.88
With crane and bucket	C.Y.	C7	80	0.9		31.55	13.84	45.39	67.46
Pile cap, 5 C.Y. to 10 C.Y., direct chute	C.Y.	C6	175	0.274		9.25	0.3	9.55	15.62
Pumped	C.Y.	C20	200	0.32		11.1	4.84	15.94	23.73
With crane and bucket	C.Y.	C7	150	0.48		16.86	7.41	24.27	35.88
Over 10 C.Y., direct chute	C.Y.	C6	215	0.223		7.51	0.25	7.76	12.73
Pumped	C.Y.	C20	240	0.267		9.25	4.03	13.28	19.72
With crane and bucket	C.Y.	C7	185	0.389		13.65	6.01	19.66	29.43
Slab on grade, up to 6" thick, direct chute	C.Y.	C6	110	0.436		14.69	0.49	15.18	25.01
Pumped	C.Y.	C20	130	0.492		17.08	7.47	24.55	36.48
With crane and bucket	C.Y.	C7	110	0.655		22.85	10.11	32.96	49.18
Over 6" thick, direct chute	C.Y.	C6	165	0.291		9.79	0.32	10.11	16.56
Pumped	C.Y.	C20	185	0.346		12.02	5.24	17.26	25.62
With crane and bucket	C.Y.	C7	145	0.497		17.41	7.67	25.08	37.23
Walls, 8" thick, direct chute	C.Y.	C6	90	0.533		17.95	0.59	18.54	30.56
Pumped	C.Y.	C20	100	0.64		22.3	9.7	32	47.62
With crane and bucket	C.Y.	C7	80	0.9		31.55	13.84	45.39	67.46
12" thick, direct chute	C.Y.	C6	100	0.48		16.16	0.53	16.69	27.24
Pumped	C.Y.	C20	110	0.582		20.18	8.81	28.99	42.88
With crane and bucket	C.Y.	C7	90	0.8		28.29	12.34	40.63	59.77
15" thick, direct chute	C.Y.	C6	105	0.457		15.4	0.51	15.91	26.12
Pumped	C.Y.	C20	120	0.533		18.5	8.09	26.59	39.33
With crane and bucket	C.Y.	C7	95	0.758		26.66	11.67	38.33	56.92
Wheelled concrete dumping, add to placing costs above									
Walking cart, 50' haul, add	C.Y.	C18	32	0.281		9.08	1.88	10.96	17.18
150' haul, add	C.Y.	C18	24	0.375		12.13	2.5	14.63	22.93
250' haul, add	C.Y.	C18	18	0.5		16.16	3.33	19.49	30.32
Riding cart, 50' haul, add	C.Y.	C19	80	0.113		3.63	1.27	4.9	7.43
150' haul, add	C.Y.	C19	60	0.15		4.85	1.68	6.53	9.91
250' haul, add	C.Y.	C19	45	0.2		6.47	2.24	8.71	13.24

Concrete in-fill for pan-type metal stairs and landings. Manual placement includes up to 50' horizontal haul from point of concrete discharge.

Stair pan treads, 2" deep									
Flights in 1st floor level up/down from discharge point	S.F.	C8A	3200	0.015		0.52		0.52	0.86
2nd floor level	S.F.	C8A	2500	0.019		0.67		0.67	1.1
3rd floor level	S.F.	C8A	2000	0.024		0.84		0.84	1.38
4th floor level	S.F.	C8A	1800	0.027		0.92		0.92	1.53
Intermediate stair landings, pan-type 4" deep									
Flights in 1st floor level up/down from discharge point	S.F.	C8A	2000	0.024		0.84		0.84	1.38
2nd floor level	S.F.	C8A	1500	0.032		1.12		1.12	1.84
3rd floor level	S.F.	C8A	1200	0.04		1.39		1.39	2.3
4th floor level	S.F.	C8A	1000	0.048		1.68		1.68	2.75

Lightweight Structural Concrete

LIGHTWEIGHT CONCRETE, READY MIX

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Lightweight, 110#/C.F.									
2500 psi	C.Y.				118.9			118.9	130.18
3000 psi	C.Y.				130.18			130.18	142.48
3500 psi	C.Y.				131.2			131.2	144.53

4000 psi	C.Y.				136.33				136.33	149.63
4500 psi	C.Y.				143.5				143.5	156.83
5000 psi	C.Y.				140.43				140.43	154.78
Concrete Finishing										
High-Tolerance Concrete Floor Finishing										
FINISHING FLOORS, HIGH TOLERANCE										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Finishing of fresh concrete flatwork requires that concrete first be placed, struck off & consolidated										
Basic finishing for various unspecified flatwork										
Bull float only	S.F.	C10	4000	0.006		0.22		0.22	0.36	
Bull float & manual float	S.F.	C10	2000	0.012		0.45		0.45	0.73	
Bull float, manual float & broom finish, w/edging & joints	S.F.	C10	1850	0.013		0.48		0.48	0.78	
Bull float, manual float & manual steel trowel	S.F.	C10	1265	0.019		0.7		0.7	1.14	
For specified Random Access Floors in ACI Classes 1, 2, 3 and 4 to achieve										
Composite Overall Floor Flatness and Levelness values up to FF35/FL25										
Bull float, machine float & machine trowel (walk-behind)	S.F.	C10C	1715	0.014		0.51	0.02	0.53	0.88	
Power screed, bull float, machine float & trowel (walk-behind)	S.F.	C10D	2400	0.01		0.37	0.05	0.42	0.65	
Power screed, bull float, machine float & trowel (ride-on)	S.F.	C10E	4000	0.006		0.22	0.06	0.28	0.43	
For specified Random Access Floors in ACI Classes 5, 6, 7 and 8 to achieve										
Composite Overall Floor Flatness and Levelness values up to FF50/FL50										
Add for two-dimensional restraightening after power float	S.F.	C10	6000	0.004		0.15		0.15	0.24	
For specified Random or Defined Access Floors in ACI Class 9 to achieve										
Composite Overall Floor Flatness and Levelness values up to FF100/FL100										
Add for two-dimensional restraightening after bull float & power float	S.F.	C10	3000	0.008		0.29		0.29	0.48	
For specified Superflat Defined Access Floors in ACI Class 9 to achieve										
Minimum Floor Flatness and Levelness values of FF100/FL100										
Add for 2-dim'l restraightening after bull float, power float, power trowel	S.F.	C10	2000	0.012		0.45		0.45	0.73	
Heavy-Duty Concrete Floor Finishing										
FINISHING FLOORS, HEAVY-DUTY										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Floor abrasives, dry shake on fresh concrete, .25 psf, aluminum oxide	S.F.	1 Cefi	850	0.009	0.56	0.37		0.93	1.23	
Silicon carbide	S.F.	1 Cefi	850	0.009	0.85	0.37		1.22	1.54	
Floor hardeners, dry shake, metallic, light service, .50 psf	S.F.	1 Cefi	850	0.009	0.44	0.37		0.81	1.08	
Medium service, .75 psf	S.F.	1 Cefi	750	0.011	0.67	0.42		1.09	1.42	
Heavy service, 1.0 psf	S.F.	1 Cefi	650	0.012	0.88	0.49		1.37	1.75	
Extra heavy, 1.5 psf	S.F.	1 Cefi	575	0.014	1.32	0.54		1.86	2.35	
Non-metallic, light service, .50 psf	S.F.	1 Cefi	850	0.009	0.16	0.37		0.53	0.78	
Medium service, .75 psf	S.F.	1 Cefi	750	0.011	0.25	0.42		0.67	0.97	
Heavy service, 1.00 psf	S.F.	1 Cefi	650	0.012	0.34	0.49		0.83	1.15	
Extra heavy, 1.50 psf	S.F.	1 Cefi	575	0.014	0.5	0.54		1.04	1.44	
Trap rock wearing surface, dry shake, for monolithic floors										
2.0 psf	S.F.	C10B	1250	0.032	0.02	1.12	0.2	1.34	2.09	
Dustproofing, liquid, for cured concrete, solvent-based, 1 coat	S.F.	1 Cefi	1900	0.004	0.16	0.16		0.32	0.45	
2 coats	S.F.	1 Cefi	1300	0.006	0.6	0.24		0.84	1.06	
Epoxy-based, 1 coat	S.F.	1 Cefi	1500	0.005	0.15	0.21		0.36	0.51	
2 coats	S.F.	1 Cefi	1500	0.005	0.31	0.21		0.52	0.68	
Colored Concrete Finishing										
FINISHING FLOORS, COLORED										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Floor coloring, dry shake on fresh concrete (0.6 psf)	S.F.	1 Cefi	1300	0.006	0.42	0.24		0.66	0.85	
(1.0 psf)	S.F.	1 Cefi	625	0.013	0.7	0.5		1.2	1.59	
Colored dry shake powder only	Lb.				0.7			0.7	0.77	
1/2" topping using 0.6 psf dry shake powdered color	S.F.	C10B	590	0.068	5.48	2.37	0.43	8.28	10.43	
1.0 psf dry shake powdered color	S.F.	C10B	590	0.068	5.74	2.37	0.43	8.54	10.74	
Exposed Aggregate Concrete Finishing										
FINISHING FLOORS, EXPOSED AGGREGATE										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Exposed local aggregate finish, seeded on fresh concrete, 3 lb./S.F.	S.F.	1 Cefi	625	0.013	0.21	0.5		0.71	1.05	
4 lb./S.F.	S.F.	1 Cefi	465	0.017	0.34	0.67		1.01	1.48	
Tooled Concrete Finishing										
FINISHING FLOORS, TOOLED										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Stair finish, fresh concrete, float finish	S.F.	1 Cefi	275	0.029		1.14		1.14	1.86	
Steel trowel finish	S.F.	1 Cefi	200	0.04		1.58		1.58	2.56	
Silicon carbide finish, dry shake on fresh concrete, .25 psf	S.F.	1 Cefi	150	0.053	0.56	2.1		2.66	4.04	
FINISHING WALLS										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Break ties and patch voids	S.F.	1 Cefi	540	0.015	0.04	0.59		0.63	1	
Burlap rub with grout	S.F.	1 Cefi	450	0.018	0.04	0.7		0.74	1.18	
Carborundum rub, dry	S.F.	1 Cefi	270	0.03		1.16		1.16	1.89	
Wet rub	S.F.	1 Cefi	175	0.046		1.8		1.8	2.92	
Bush hammer, green concrete	S.F.	B39	1000	0.048		1.56	0.24	1.8	2.86	
Cured concrete	S.F.	B39	650	0.074		2.39	0.37	2.76	4.39	
Acid etch	S.F.	1 Cefi	575	0.014	0.13	0.54		0.67	1.04	
Float finish, 1/16" thick	S.F.	1 Cefi	300	0.027	0.4	1.04		1.44	2.13	
Sandblast, light penetration	S.F.	E11	1100	0.029	0.56	1.06	0.19	1.81	2.66	
Heavy penetration	S.F.	E11	375	0.085	1.12	3.09	0.56	4.77	7.19	
Grind form fins flush	L.F.	1 Clab	700	0.011		0.37		0.37	0.61	
Stamped Concrete Finishing										
SLAB TEXTURE STAMPING										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Stamping requires that concrete first be placed, struck off, consolidated, bull floated and free of bleed water. Decorative stamping tasks include:										
Step 1 - first application of dry shake colored hardener	S.F.	1 Cefi	6400	0.001	0.44	0.05		0.49	0.56	
Step 2 - bull float	S.F.	1 Cefi	6400	0.001		0.05		0.05	0.08	
Step 3 - second application of dry shake colored hardener	S.F.	1 Cefi	6400	0.001	0.22	0.05		0.27	0.32	
Step 4 - bull float, manual float & steel trowel	S.F.	3 Cefi	1280	0.019		0.74		0.74	1.2	
Step 5 - application of dry shake colored release agent	S.F.	1 Cefi	6400	0.001	0.1	0.05		0.15	0.19	
Step 6 - place, tamp & remove mats	S.F.	3 Cefi	2400	0.01	0.83	0.39		1.22	1.55	
Step 7 - touch up edges, mat joints & simulated grout lines	S.F.	1 Cefi	1280	0.006		0.25		0.25	0.4	
Alternate stamping estimating method includes all tasks above	S.F.	4 Cefi	800	0.04	1.59	1.58		3.17	4.3	
Step 8 - pressure wash @ 3000 psi after 24 hours	S.F.	1 Cefi	1600	0.005		0.2		0.2	0.32	
Step 9 - roll 2 coats cure/seal compound when dry	S.F.	1 Cefi	800	0.01	0.65	0.39		1.04	1.35	
Polished Concrete Finishing										
POLISHED CONCRETE FLOORS										
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P	
Processing of cured concrete to include grinding, honing, and polishing of interior floors with 22" segmented diamond planetary floor grinder (2 passes in different directions per grit)										
Removal of pre-existing coatings, dry, with carbide discs using										

dry vacuum pick-up system, final hand sweeping									
Glue, adhesive or tar	M.S.F.	J4	1.6	15	21.53	554.88	157.62	734.03	1099.8
Paint, epoxy, 1 coat	M.S.F.	J4	3.6	6.667	21.53	245.89	70	337.42	503.4
2 coats	M.S.F.	J4	1.8	13.333	21.53	489.6	140	651.13	983.21
Grinding and edging, wet, including wet vac pick-up and auto scrubbing between grit changes									
40-grit diamond/metal matrix	M.S.F.	J4A	1.6	20	42.54	712.64	321.47	1076.65	1568.82
80-grit diamond/metal matrix	M.S.F.	J4A	2	16	42.54	571.2	258.21	871.95	1266.46
120-grit diamond/metal matrix	M.S.F.	J4A	2.4	13.333	42.54	478.72	214.66	735.92	1066.44
200-grit diamond/metal matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
Spray on dye or stain (1 coat)	M.S.F.	1 Cefl	16	0.5	227.55	19.69		247.24	282.2
Spray on densifier/hardener (2 coats)	M.S.F.	1 Cefl	8	1	374.13	39.17		413.3	473.65
Auto scrubbing after 2nd coat, when dry	M.S.F.	J4B	16	0.5		16.05	16.49	32.54	44.81
Honing and edging, wet, including wet vac pick-up and auto scrubbing between grit changes									
100-grit diamond/resin matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
200-grit diamond/resin matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
Dry, including dry vacuum pick-up system, final hand sweeping									
400-grit diamond/resin matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
Polishing and edging, dry, including dry vac pick-up and hand sweeping between grit changes									
800-grit diamond/resin matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
1500-grit diamond/resin matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
3000-grit diamond/resin matrix	M.S.F.	J4A	2.8	11.429	42.54	408	184.59	635.13	919.01
Auto scrubbing after final polishing step	M.S.F.	J4B	16	0.5		16.05	16.49	32.54	44.81

Specialty Placed Concrete

Shotcrete

GUNITED (DRY-MIX)

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Typical in place, 1" layers, no mesh included	S.F.	C16	2000	0.028	0.37	1.01	0.2	1.58	2.29
Mesh for gunite 2 x 2, #12	S.F.	2 Rodm	800	0.02	0.72	0.88		1.6	2.26
#4 reinforcing bars @ 6" each way	S.F.	2 Rodm	500	0.032	1.64	1.4		3.04	4.15
Typical in place, including mesh, 2" thick, flat surfaces	S.F.	C16	1000	0.056	1.47	2.02	0.4	3.89	5.39
Curved surfaces	S.F.	C16	500	0.112	1.47	4.04	0.81	6.32	9.13
4" thick, flat surfaces	S.F.	C16	750	0.075	2.21	2.7	0.54	5.45	7.46
Curved surfaces	S.F.	C16	350	0.16	2.21	5.77	1.15	9.13	13.22
Prepare old walls, no scaffolding, good condition	S.F.	C10	1000	0.024		0.88		0.88	1.45
Poor condition	S.F.	C10	275	0.087		3.22		3.22	5.27
For high finish requirement or close tolerance, add	S.F.					50.00%			
Very high	S.F.					110.00%			

SHOTCRETE (WET-MIX)

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Wet mix, placed @ up to 12 C.Y./hour, 3000 psi	C.Y.	C8C	80	0.6	120.95	21.32	6.48	148.75	175.77
Up to 35 C.Y./hour	C.Y.	C8E	240	0.2	108.65	7.07	2.49	118.21	134.31
Fiber reinforced, 1" thick	S.F.	C8C	1740	0.028	0.89	0.98	0.3	2.17	2.92
2" thick	S.F.	C8C	900	0.053	1.78	1.89	0.58	4.25	5.72
3" thick	S.F.	C8C	825	0.058	2.67	2.07	0.63	5.37	7.04
4" thick	S.F.	C8C	750	0.064	3.56	2.27	0.69	6.52	8.44

Roller-Compacted Concrete

CONCRETE, ROLLER-COMPACTED

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Mass placement, 1' lift, 1' layer	C.Y.	B10C	1280	0.006		0.27	1.38	1.65	1.96
2' lift, 6" layer	C.Y.	B10C	1600	0.005		0.22	1.1	1.32	1.57
Vertical face, formed, 1' lift	C.Y.	B11V	400	0.06		1.93	0.47	2.4	3.71
6" lift	C.Y.	B11V	200	0.12		3.85	0.93	4.78	7.45
Sloped face, nonformed, 1' lift	C.Y.	B11L	384	0.042		1.57	1.74	3.31	4.51
6" lift	C.Y.	B11L	192	0.083		3.14	3.47	6.61	9.03
Surface preparation, vacuum truck	S.Y.	B6A	3280	0.006		0.23	0.11	0.34	0.49
Water clean	S.Y.	B9A	3000	0.008		0.27	0.16	0.43	0.62
Water blast	S.Y.	B9B	800	0.03		1.01	0.65	1.66	2.4
Joint bedding placement, 1" thick	S.Y.	B11C	975	0.016		0.62	0.33	0.95	1.38
Conveyance of materials, 18 C.Y. truck, 5 min. cycle	C.Y.	B34F	2048	0.004		0.14	0.74	0.88	1.05
10 min. cycle	C.Y.	B34F	1024	0.008		0.29	1.46	1.75	2.1
15 min. cycle	C.Y.	B34F	680	0.012		0.44	2.21	2.65	3.15
With crane and bucket	C.Y.	C23A	1600	0.025		0.9	1.62	2.52	3.28
With 4 C.Y. loader, 4 min. cycle	C.Y.	B10U	480	0.017		0.72	2.13	2.85	3.52
8 min. cycle	C.Y.	B10U	240	0.033		1.45	4.24	5.69	7.05
12 min. cycle	C.Y.	B10U	160	0.05		2.17	6.38	8.55	10.57
With belt conveyor	C.Y.	C7D	600	0.093		3.18	0.32	3.5	5.62
With 17 C.Y. scraper, 5 min. cycle	C.Y.	B33J	1440	0.006		0.24	1.79	2.03	2.36
10 min. cycle	C.Y.	B33J	720	0.011		0.48	3.59	4.07	4.73
15 min. cycle	C.Y.	B33J	480	0.017		0.72	5.39	6.11	7.1
20 min. cycle	C.Y.	B33J	360	0.022		0.96	7.16	8.12	9.47
Water cure, small job, \$L 500 C.Y.	Hr.	B94C	8	1		32.1	11.04	43.14	65.44
Large job, over 500 C.Y.	Hr.	B59A	8	3		101.18	56.52	157.7	229.77
RCC paving, with asphalt paver including material	C.Y.	B25C	1000	0.048	75.85	1.74	2.44	80.03	89.11
8" thick layers	S.Y.	B25C	4200	0.011	19.78	0.41	0.58	20.77	23.37
12" thick layers	S.Y.	B25C	2800	0.017	29.21	0.62	0.87	30.7	34.26
Roller compacted concrete, 1.5" - 2" agg., 200 lb. cement/C.Y.	C.Y.				89.18			89.18	97.89

Concrete Curing

Water Concrete Curing

WATER CURING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
With burlap, 4 uses assumed, 7.5 oz.	C.S.F.	2 Clab	55	0.291	15.12	9.36		24.48	32.11
10 oz.	C.S.F.	2 Clab	55	0.291	27.16	9.36		36.52	45.23
Curing blankets, 1" to 2" thick, buy	S.F.				0.27			0.27	0.3

Membrane Concrete Curing

CHEMICAL COMPOUND MEMBRANE CONCRETE CURING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Sprayed membrane curing compound	C.S.F.	2 Clab	95	0.168	12.4	5.41		17.81	22.66
Curing compound, solvent based, 400 S.F./gal., 55 gallon lots	Gal.				22.04			22.04	24.09
5 gallon lots	Gal.				32.8			32.8	35.88
Curing compound, water based, 250 S.F./gal., 55 gallon lots	Gal.				22.04			22.04	24.09
5 gallon lots	Gal.				24.6			24.6	27.16

SHEET MEMBRANE CONCRETE CURING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Curing blanket, burlap/poly, 2-ply	C.S.F.	2 Clab	70	0.229	25.63	7.34		32.97	40.38

Precast Structural Concrete

Precast Concrete Members

Precast Concrete Hollow Core Planks

PRECAST SLAB PLANKS

Prestressed roof/floor members, grouted, solid, 4" thick	S.F.	C11	2400	0.023	6.7	0.91	0.87	8.48	9.83
6" thick	S.F.	C11	2800	0.02	7.5	0.78	0.75	9.03	10.37
Hollow, 8" thick	S.F.	C11	3200	0.018	8.38	0.68	0.65	9.71	11.08
10" thick	S.F.	C11	3600	0.016	8.67	0.6	0.58	9.85	11.2
12" thick	S.F.	C11	4000	0.014	9.01	0.54	0.52	10.07	11.37

Precast Concrete Slabs

PRECAST CONCRETE CHANNEL SLABS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Lightweight concrete channel slab, long runs, 2-3/4" thick	S.F.	C12	1575	0.03	9.51	1.13	0.32	10.96	12.71
3-3/4" thick	S.F.	C12	1550	0.031	9.8	1.15	0.33	11.28	13.03
4-3/4" thick	S.F.	C12	1525	0.031	10.89	1.17	0.34	12.4	14.29
Short pieces, 2-3/4" thick	S.F.	C12	785	0.061	14.29	2.27	0.65	17.21	20.2
3-3/4" thick	S.F.	C12	770	0.062	14.67	2.31	0.66	17.64	20.7
4-3/4" thick	S.F.	C12	762	0.063	16.38	2.34	0.67	19.39	22.64

PRECAST LIGHTWEIGHT CONCRETE PLANK

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Lightweight plank, nailable, T&G, 2" thick	S.F.	C12	1800	0.027	8.46	0.99	0.29	9.74	11.25
For premium ceiling finish, add	S.F.				10.00%				
For sloping roofs, slope over 4 in 12, add						25.00%			
Slope over 6 in 12, add						150.00%			

Precast Concrete Stairs

PRECAST STAIRS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Precast concrete treads on steel stringers, 3' wide	Riser	C12	75	0.64	128.21	23.76	6.84	158.81	187.95
Front entrance 4' wide with 48" platform, 2 risers	Flight	C12	17	2.824	460.9	104.94	30.07	595.91	713.42
5 risers	Flight	C12	13	3.692	708.11	136.62	39.41	884.14	1049.6
Front entrance, 5' wide with 48" platform, 2 risers	Flight	C12	16	3	523.75	110.88	32.15	666.78	797.62
5 risers	Flight	C12	12	4	833.81	148.5	43.04	1025.35	1214.5
6' wide, 2 risers	Flight	C12	15	3.2	582.41	118.8	34.22	735.43	875.93
5 risers	Flight	C12	11	4.364	921.8	161.37	46.67	1129.84	1325.22
7' wide, 2 risers	Flight	C12	14	3.429	741.63	126.72	36.81	905.16	1068.36
5 risers	Flight	C12	10	4.8	1236.05	178.2	51.33	1465.58	1713.29
Basement entrance stairwell, 6 steps, incl. steel bulkhead door	Flight	B51	22	2.182	1571.25	65.84	8.87	1645.96	1836.6
14 steps	Flight	B51	11	4.364	2618.75	131.67	17.78	2768.2	3108.49

Precast Structural Pretensioned Concrete

PRECAST BEAMS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
L-shaped, 20' span, 12" x 20"	Ea.	C11	32	1.75	3435.8	67.82	65.33	3568.95	3977.87
18" x 36"	Ea.	C11	24	2.333	4692.8	90.59	87.11	4870.5	5422.04
24" x 44"	Ea.	C11	22	2.545	5635.55	99	94.89	5829.44	6471.27
30' span, 12" x 36"	Ea.	C11	24	2.333	6347.85	90.59	87.11	6525.55	7223.74
18" x 44"	Ea.	C11	20	2.8	7730.55	108.9	104.74	7944.19	8843.88
24" x 52"	Ea.	C11	16	3.5	9301.8	135.63	130.66	9568.09	10594.45
40' span, 12" x 52"	Ea.	C11	20	2.8	9888.4	108.9	104.74	10102.04	11106.48
18" x 52"	Ea.	C11	16	3.5	10977.8	135.63	130.66	11244.09	12438.05
24" x 52"	Ea.	C11	12	4.667	12402.4	181.17	174.22	12757.79	14069.4
Rectangular, 20' span, 12" x 20"	Ea.	C11	32	1.75	3331.05	67.82	65.33	3464.2	3852.17
18" x 36"	Ea.	C11	24	2.333	4127.15	90.59	87.11	4304.85	4793.54
24" x 44"	Ea.	C11	22	2.545	4923.25	99	94.89	5117.14	5696.12
30' span, 12" x 36"	Ea.	C11	24	2.333	5551.75	90.59	87.11	5729.45	6364.79
18" x 44"	Ea.	C11	20	2.8	6704	108.9	104.74	6917.64	7670.68
24" x 52"	Ea.	C11	16	3.5	8065.75	135.63	130.66	8332.04	9253.65
40' span, 12" x 52"	Ea.	C11	20	2.8	8254.3	108.9	104.74	8467.94	9346.68
18" x 52"	Ea.	C11	16	3.5	9385.6	135.63	130.66	9651.89	10678.25
24" x 52"	Ea.	C11	12	4.667	10726.4	181.17	174.22	11081.79	12309.6
T" shaped, 20' span, 12" x 20"	Ea.	C11	32	1.75	3917.65	67.82	65.33	4050.8	4480.67
18" x 36"	Ea.	C11	24	2.333	5258.45	90.59	87.11	5436.15	6029.59
24" x 44"	Ea.	C11	22	2.545	6305.95	99	94.89	6499.84	7204.52
30' span, 12" x 36"	Ea.	C11	24	2.333	7227.75	90.59	87.11	7405.45	8187.44
18" x 44"	Ea.	C11	20	2.8	8799	108.9	104.74	9012.64	9933.28
24" x 52"	Ea.	C11	16	3.5	10475	135.63	130.66	10741.29	11935.25
40' span, 12" x 52"	Ea.	C11	20	2.8	11899.6	108.9	104.74	12113.24	13369.08
18" x 52"	Ea.	C11	16	3.5	12653.8	135.63	130.66	12920.09	14281.65
24" x 52"	Ea.	C11	12	4.667	13994.6	181.17	174.22	14349.99	15829.2

PRECAST COLUMNS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Rectangular to 12' high, 16" x 16"	L.F.	C11	120	0.467	196.09	18.12	17.42	231.63	264.75
24" x 24"	L.F.	C11	96	0.583	263.97	22.77	21.78	308.52	350.58
24' high, 28" x 28"	L.F.	C11	192	0.292	305.87	11.34	10.89	328.1	366.09
36" x 36"	L.F.	C11	144	0.389	410.62	15.1	14.52	440.24	493.74
24' high, 1 haunch, 12" x 12"	Ea.	C11	32	1.75	4148.1	67.82	65.33	4281.25	4753.02
20" x 20"	Ea.	C11	28	2	5677.45	77.72	74.66	5829.83	6475.66

PRECAST JOISTS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
40 psf L.L., 6" deep for 12' spans	L.F.	C12	600	0.08	27.65	2.96	0.86	31.47	36.03
8" deep for 16' spans	L.F.	C12	575	0.083	45.67	3.09	0.89	49.65	56.42
10" deep for 20' spans	L.F.	C12	550	0.087	80.03	3.23	0.93	84.19	94.37
12" deep for 24' spans	L.F.	C12	525	0.091	109.78	3.39	0.97	114.14	127.39

PRECAST TEES

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Quad tee, short spans, roof	S.F.	C11	7200	0.008	8.72	0.31	0.29	9.32	10.42
Floor	S.F.	C11	7200	0.008	8.72	0.31	0.29	9.32	10.42
Double tee, floor members, 60' span	S.F.	C11	8400	0.007	9.8	0.26	0.25	10.31	11.52
80' span	S.F.	C11	8000	0.007	13.58	0.27	0.26	14.11	15.71
Roof members, 30' span	S.F.	C11	4800	0.012	9.43	0.46	0.44	10.33	11.58
50' span	S.F.	C11	6400	0.009	9.6	0.34	0.33	10.27	11.44
Wall members, up to 55' high	S.F.	C11	3600	0.016	13.03	0.6	0.58	14.21	15.98
Single tee roof members, 40' span	S.F.	C11	3200	0.018	13.07	0.68	0.65	14.4	16.27
80' span	S.F.	C11	5120	0.011	13.49	0.43	0.4	14.32	15.99
100' span	S.F.	C11	6000	0.009	20.11	0.37	0.35	20.83	23.19
120' span	S.F.	C11	6000	0.009	21.79	0.37	0.35	22.51	24.86

Double tees, floor members

Lightweight, 20" x 8' wide, 45' span	Ea.	C11	20	2.8	3456.75	108.9	104.74	3670.39	4088.23
24" x 8' wide, 50' span	Ea.	C11	18	3.111	3833.85	120.78	116.14	4070.77	4540.46
32" x 10' wide, 60' span	Ea.	C11	16	3.5	5761.25	135.63	130.66	6027.54	6697.75
Standard weight, 12" x 8' wide, 20' span	Ea.	C11	22	2.545	1403.65	99	94.89	1597.54	1799.42
16" x 8' wide, 25' span	Ea.	C11	20	2.8	1738.85	108.9	104.74	1952.49	2223.68
18" x 8' wide, 30' span	Ea.	C11	20	2.8	2095	108.9	104.74	2308.64	2600.78
20" x 8' wide, 45' span	Ea.	C11	18	3.111	3142.5	120.78	116.14	3379.42	3786.26
24" x 8' wide, 50' span	Ea.	C11	16	3.5	3477.7	135.63	130.66	3743.99	4204.7

32" x 10' wide, 60' span	Ea.	C11	14	4	5237.5	155.43	149.33	5542.26	6184.48
Roof members									
Lightweight, 20" x 8' wide, 40' span	Ea.	C11	20	2.8	3058.7	108.9	104.74	3272.34	3669.23
24" x 8' wide, 50' span	Ea.	C11	18	3.111	3833.85	120.78	116.14	4070.77	4540.46
32" x 10' wide, 60' span	Ea.	C11	16	3.5	5761.25	135.63	130.66	6027.54	6697.75
Standard weight, 12" x 8' wide, 30' span	Ea.	C11	22	2.545	2095	99	94.89	2288.89	2574.57
16" x 8' wide, 30' span	Ea.	C11	20	2.8	2199.75	108.9	104.74	2413.39	2705.53
18" x 8' wide, 30' span	Ea.	C11	20	2.8	2304.5	108.9	104.74	2518.14	2831.23
20" x 8' wide, 40' span	Ea.	C11	18	3.111	2786.35	120.78	116.14	3023.27	3388.21
24" x 8' wide, 50' span	Ea.	C11	16	3.5	3477.7	135.63	130.66	3743.99	4204.7
32" x 10' wide, 60' span	Ea.	C11	14	4	5237.5	155.43	149.33	5542.26	6184.48

Precast Architectural Concrete

Faced Architectural Precast Concrete

PRECAST WALL PANELS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Uninsulated, smooth gray									
Low rise, 4' x 8' x 4" thick	S.F.	C11	320	0.175	23.05	6.78	6.53	36.36	43.69
8' x 8', 4" thick	S.F.	C11	576	0.097	22.63	3.78	3.63	30.04	35.42
8' x 16' x 4" thick	S.F.	C11	1024	0.055	22.63	2.13	2.04	26.8	30.51
High rise, 4' x 8' x 4" thick	S.F.	C11	288	0.194	23.05	7.57	7.26	37.88	45.74
8' x 8' x 4" thick	S.F.	C11	512	0.109	22.63	4.25	4.09	30.97	36.71
8' x 16' x 4" thick	S.F.	C11	768	0.073	22.63	2.83	2.73	28.19	32.45
10' x 20', 6" thick	S.F.	C11	1400	0.04	38.55	1.55	1.49	41.59	46.55
Insulated panel, 2" polystyrene, add	S.F.				0.96			0.96	1.05
2" urethane, add	S.F.				0.75			0.75	0.83
Finishes, white, add	S.F.				2.77			2.77	3.05
Exposed aggregate, add	S.F.				0.58			0.58	0.64
Granite faced, domestic, add	S.F.				25.56			25.56	28.07
Brick faced, modular, red, add	S.F.				3.91			3.91	4.32
Fiberglass reinforced cement with urethane core									
R20, 8' x 8', 5" plain finish	S.F.	E2	750	0.064	23.46	2.59	2.33	28.38	33.08
Exposed aggregate or brick finish	S.F.	E2	600	0.08	35.62	3.24	2.91	41.77	47.86

Site-Cast Concrete

Tilt-Up Concrete

TILT-UP WALL PANELS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Wall panel construction, walls only, 5-1/2" thick	S.F.	C14	1600	0.085	4.99	2.87	0.99	8.85	11.33
7-1/2" thick	S.F.	C14	1550	0.088	6.2	2.96	1.02	10.18	12.82
Walls and columns, 5-1/2" thick walls, 12" x 12" columns	S.F.	C14	1565	0.087	7.33	2.93	1.01	11.27	14.07
7-1/2" thick wall, 12" x 12" columns	S.F.	C14	1370	0.099	8.92	3.35	1.15	13.42	16.61
Columns only, site precast, 12" x 12"	L.F.	C14	200	0.68	17.18	22.77	7.88	47.83	65.64
16" x 16"	L.F.	C14	105	1.295	25.14	43.56	14.98	83.68	116.5

Precast Concrete Specialties

Precast Concrete Trim

PRECAST LINTELS, smooth gray, prestressed, stock units only

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
4" wide, 8" high, x 4' long	Ea.	D10	28	1.143	27.24	41.09	13.48	81.81	113.31
8' long	Ea.	D10	24	1.333	68.72	48.02	15.71	132.45	172.39
6" wide, 8" high, x 4' long	Ea.	D10	26	1.231	40.22	44.06	14.52	98.8	134.14
10' long	Ea.	D10	22	1.455	100.56	52.47	17.11	170.14	216.61
8" wide, 8" high, x 4' long	Ea.	D10	24	1.333	47.35	48.02	15.71	111.08	148.93
12' long	Ea.	D10	20	1.6	153.35	57.42	18.87	229.64	285.56
For custom sizes, types, colors, or finishes of precast lintels, add	Ea.				150.00%				

PRECAST WINDOW SILLS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Precast concrete, 4" tapers to 3", 9" wide	L.F.	D1	70	0.229	15.92	7.47		23.39	30.07
11" wide	L.F.	D1	60	0.267	25.98	8.71		34.69	43.04

Cast Roof Decks

Cementitious Wood Fiber Decks

CEMENTITIOUS/WOOD FIBER PLANKS

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Plank, beveled edge, 1" thick	S.F.	2 Carp	1000	0.016	3.35	0.65		4	4.75
1-1/2" thick	S.F.	2 Carp	975	0.016	4.34	0.66		5	5.87
T&G, 2" thick	S.F.	2 Carp	950	0.017	3.68	0.68		4.36	5.19
2-1/2" thick	S.F.	2 Carp	925	0.017	4.02	0.7		4.72	5.58
3" thick	S.F.	2 Carp	900	0.018	4.55	0.72		5.27	6.2
Bulb tee, sub-purlin and grout, 6' span, add	S.F.	E1	5000	0.003	2.12	0.14	0.02	2.28	2.6
8' span	S.F.	E1	4200	0.004	2.12	0.16	0.02	2.3	2.66

Gypsum Concrete Roof Decks

GYPSUM ROOF DECK

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Poured gypsum, 2" thick	S.F.	C8	6000	0.009	1.71	0.33	0.16	2.2	2.59
3" thick	S.F.	C8	4800	0.012	2.56	0.41	0.19	3.16	3.71

Lightweight Concrete Roof Insulation

Lightweight Insulating Concrete

LIGHTWEIGHT CELLULAR INSULATING CONCRETE

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Portland cement and foaming agent	C.Y.	C8	50	1.12	125.7	39.29	18.3	183.29	222.38

LIGHTWEIGHT AGGREGATE INSULATING CONCRETE

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Poured vermiculite or perlite, field mix,									
1:6 field mix	C.Y.	C8	50	1.12	250.41	39.29	18.3	308	360.84
Ready mix, 1:6 mix, roof fill, 2" thick	S.F.	C8	10000	0.006	1.39	0.2	0.09	1.68	1.96
3" thick	S.F.	C8	7700	0.007	2.09	0.25	0.11	2.45	2.85
Expanded volcanic glass rock, 1" thick	S.F.	2 Carp	1500	0.011	0.61	0.44		1.05	1.38
3" thick	S.F.	2 Carp	1200	0.013	1.84	0.54		2.38	2.9
Lightweight insulating fill									
1000 psi	C.Y.				110.97			110.97	121.77
1:6 mix perlite	C.Y.				112.93			112.93	124.71

Concrete Topping

Iron-Aggregate Concrete Topping

FLOOR TOPPING

Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Integral topping/finish, on fresh concrete, using 1:1:2 mix, 3/16" thick	S.F.	C10B	1000	0.04	0.12	1.37	0.25	1.74	2.67
1/2" thick	S.F.	C10B	950	0.042	0.32	1.43	0.27	2.02	3.01
3/4" thick	S.F.	C10B	850	0.047	0.48	1.6	0.3	2.38	3.49
1" thick	S.F.	C10B	750	0.053	0.65	1.83	0.33	2.81	4.07
Granolithic topping, on fresh or cured concrete, 1:1:1-1/2 mix, 1/2" thick	S.F.	C10B	590	0.068	0.35	2.32	0.43	3.1	4.67
3/4" thick	S.F.	C10B	580	0.069	0.54	2.36	0.44	3.34	4.95
1" thick	S.F.	C10B	575	0.07	0.72	2.38	0.44	3.54	5.19
2" thick	S.F.	C10B	500	0.08	1.43	2.73	0.51	4.67	6.62

Heavy duty, 1:1:2, 3/4" thick, preshrunk, gray, 20 M.S.F.	S.F.	C10B	320	0.125	0.87	4.27	0.79	5.93	8.84
100 M.S.F.	S.F.	C10B	380	0.105	0.48	3.6	0.66	4.74	7.15
Cast Underlayment									
Gypsum Cement Underlayment									
POURED GYPSUM UNDERLAYMENT									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Underlayment, gypsum based, self-leveling 2500 psi, pumped, 1/2" thick	S.F.	C8	24000	0.002	0.42	0.08	0.04	0.54	0.65
3/4" thick	S.F.	C8	20000	0.003	0.64	0.1	0.04	0.78	0.92
1" thick	S.F.	C8	16000	0.004	0.85	0.13	0.06	1.04	1.2
Hand placed, 1/2" thick	S.F.	C18	450	0.02	0.42	0.63	0.13	1.18	1.67
3/4" thick	S.F.	C18	300	0.03	0.64	0.95	0.2	1.79	2.5
Hydraulic Cement Underlayment									
CEMENT UNDERLAYMENT									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Underlayment, P.C. based, self-leveling, 4100 psi, pumped, 1/4" thick	S.F.	C8	20000	0.003	1.65	0.1	0.04	1.79	2.03
1/2" thick	S.F.	C8	19000	0.003	3.3	0.11	0.05	3.46	3.84
3/4" thick	S.F.	C8	18000	0.003	4.96	0.11	0.05	5.12	5.68
1" thick	S.F.	C8	17000	0.003	6.58	0.12	0.05	6.75	7.52
1-1/2" thick	S.F.	C8	15000	0.004	9.92	0.13	0.06	10.11	11.17
Hand placed, 1/2" thick	S.F.	C18	450	0.02	3.3	0.63	0.13	4.06	4.82
POURED PORTLAND CEMENT TOPPING									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Topping, P.C. based, self-leveling, 6100 psi, pumped, 1/4" thick	S.F.	C8	20000	0.003	2.43	0.1	0.04	2.57	2.87
1/2" thick	S.F.	C8	19000	0.003	4.84	0.11	0.05	5	5.57
3/4" thick	S.F.	C8	18000	0.003	7.27	0.11	0.05	7.43	8.23
1" thick	S.F.	C8	17000	0.003	9.67	0.12	0.05	9.84	10.9
1-1/2" thick	S.F.	C8	15000	0.004	14.53	0.13	0.06	14.72	16.28
Hand placed, 1/2" thick	S.F.	C18	450	0.02	4.84	0.63	0.13	5.6	6.55
Cementitious Grouting									
Dry-Pack Grouting									
GROUT, DRY-PACK									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Dry-pack, under beams or walls, cement & sand, 1" thick	S.F.	D3	170	0.247	0.73	8.87		9.6	15.68
2" thick	S.F.	D3	150	0.28	1.45	10.09		11.54	18.43
Non-Shrink Grouting									
Non-Metallic Non-Shrink Grouting									
GROUT, NON-METALLIC NON-SHRINK									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Non-shrink, non-metallic, 1" deep	S.F.	1 Cefl	35	0.229	6.82	8.76		15.58	21.74
2" deep	S.F.	1 Cefl	25	0.32	13.6	12.27		25.87	34.89
Metallic Non-Shrink Grouting									
GROUT, METALLIC NON-SHRINK									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Column & machine bases, non-shrink, metallic, 1" deep	S.F.	1 Cefl	35	0.229	11.93	8.76		20.69	27.39
2" deep	S.F.	1 Cefl	25	0.32	24.06	12.27		36.33	46.42
Epoxy Grouting									
Grouting of Dowels and Fasteners									
EPOXY ONLY									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Chemical anchoring, epoxy cartridge, excludes layout, drilling, fastener									
For fastener 3/4" diam. x 6" embedment	Ea.	2 Skwk	72	0.222	5.11	9.13		14.24	20.89
1" diam. x 8" embedment	Ea.	2 Skwk	66	0.242	7.66	9.93		17.59	25.07
1-1/4" diam. x 10" embedment	Ea.	2 Skwk	60	0.267	15.37	10.94		26.31	35.21
1-3/4" diam. x 12" embedment	Ea.	2 Skwk	54	0.296	25.53	12.16		37.69	48.33
14" embedment	Ea.	2 Skwk	48	0.333	30.93	13.7		44.63	56.71
2" diam. x 12" embedment	Ea.	2 Skwk	42	0.381	40.75	15.61		56.36	71.19
18" embedment	Ea.	2 Skwk	32	0.5	51.06	20.55		71.61	90.99
Injection Grouting									
Epoxy Injection Grouting									
Concrete Cutting									
Flat Concrete Sawing									
CONCRETE FLOOR/SLAB CUTTING									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes blade cost, layout and set-up time									
Saw cut concrete slabs, plain, up to 3" deep	L.F.	B89	1060	0.015	0.12	0.54	0.41	1.07	1.51
Each additional inch of depth	L.F.	B89	3180	0.005	0.04	0.18	0.13	0.35	0.52
Mesh reinforced, up to 3" deep	L.F.	B89	980	0.016	0.14	0.59	0.45	1.18	1.63
Each additional inch of depth	L.F.	B89	2940	0.005	0.05	0.2	0.15	0.4	0.55
Rod reinforced, up to 3" deep	L.F.	B89	800	0.02	0.17	0.72	0.55	1.44	2
Each additional inch of depth	L.F.	B89	2400	0.007	0.06	0.24	0.19	0.49	0.66
CONCRETE SAW BLADES									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Blades for saw cutting, included in cutting line items									
Diamond, 12" diameter	Ea.				216.04			216.04	237.64
18" diameter	Ea.				407.53			407.53	451.72
24" diameter	Ea.				682.49			682.49	751.23
30" diameter	Ea.				962.36			962.36	1055.65
36" diameter	Ea.				1276.6			1276.6	1399.35
42" diameter	Ea.				2504.1			2504.1	2749.6
Track Mounted Concrete Wall Sawing									
CONCRETE WALL CUTTING									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes blade cost, layout and set-up time									
Concrete walls, hydraulic saw, plain, per inch of depth	L.F.	B89B	250	0.064	0.04	2.44	2.79	5.27	7.13
Rod reinforcing, per inch of depth	L.F.	B89B	150	0.107	0.06	4.07	4.65	8.78	11.86
Concrete Boring									
Concrete Core Drilling									
CORE DRILLING									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes bit cost, layout and set-up time									
Reinforced concrete slab, up to 6" thick									
1" diameter core	Ea.	B89A	17	0.941	0.18	33.98	6.84	41	64.54
For each additional inch of slab thickness in same hole, add	Ea.	B89A	1440	0.011	0.03	0.4	0.08	0.51	0.79
2" diameter core	Ea.	B89A	16.5	0.97	0.28	35.05	7.05	42.38	66.5
For each additional inch of slab thickness in same hole, add	Ea.	B89A	1080	0.015	0.05	0.53	0.1	0.68	1.05
3" diameter core	Ea.	B89A	16	1	0.38	36.11	7.26	43.75	68.93
For each additional inch of slab thickness in same hole, add	Ea.	B89A	720	0.022	0.07	0.81	0.17	1.05	1.59
4" diameter core	Ea.	B89A	15	1.067	0.49	38.76	7.78	47.03	73.29
For each additional inch of slab thickness in same hole, add	Ea.	B89A	480	0.033	0.08	1.21	0.24	1.53	2.37
6" diameter core	Ea.	B89A	14	1.143	0.76	41.42	8.3	50.48	78.99
For each additional inch of slab thickness in same hole, add	Ea.	B89A	360	0.044	0.13	1.6	0.32	2.05	3.18

8" diameter core	Ea.	B89A	13	1.231	1.16	44.6	8.97	54.73	85.47
For each additional inch of slab thickness in same hole, add	Ea.	B89A	288	0.056	0.2	2.01	0.4	2.61	4.03
10" diameter core	Ea.	B89A	12	1.333	1.46	48.32	9.7	59.48	93
For each additional inch of slab thickness in same hole, add	Ea.	B89A	240	0.067	0.25	2.41	0.49	3.15	4.82
12" diameter core	Ea.	B89A	11	1.455	1.8	52.57	10.58	64.95	101.2
For each additional inch of slab thickness in same hole, add	Ea.	B89A	206	0.078	0.29	2.81	0.56	3.66	5.63
14" diameter core	Ea.	B89A	10	1.6	2.08	57.88	11.61	71.57	111.74
For each additional inch of slab thickness in same hole, add	Ea.	B89A	180	0.089	0.34	3.22	0.64	4.2	6.46
18" diameter core	Ea.	B89A	9	1.778	2.87	64.25	12.91	80.03	124.63
For each additional inch of slab thickness in same hole, add	Ea.	B89A	144	0.111	0.48	4.02	0.81	5.31	8.11
24" diameter core	Ea.	B89A	8	2	4.07	72.22	14.52	90.81	141.51
For each additional inch of slab thickness in same hole, add	Ea.	B89A	120	0.133	0.68	4.82	0.96	6.46	9.89
For horizontal holes, add to above	Ea.					20.00%	20.00%		
Prestressed hollow core plank, 8" thick									
1" diameter core	Ea.	B89A	17.5	0.914	0.24	32.92	6.64	39.8	62.8
For each additional inch of plank thickness in same hole, add	Ea.	B89A	3840	0.004	0.03	0.15	0.03	0.21	0.31
2" diameter core	Ea.	B89A	17.25	0.928	0.38	33.45	6.74	40.57	63.58
For each additional inch of plank thickness in same hole, add	Ea.	B89A	2880	0.006	0.05	0.2	0.04	0.29	0.43
3" diameter core	Ea.	B89A	17	0.941	0.51	33.98	6.84	41.33	64.9
For each additional inch of plank thickness in same hole, add	Ea.	B89A	1920	0.008	0.07	0.3	0.06	0.43	0.63
4" diameter core	Ea.	B89A	16.5	0.97	0.66	35.05	7.05	42.76	66.92
For each additional inch of plank thickness in same hole, add	Ea.	B89A	1280	0.013	0.08	0.46	0.09	0.63	0.94
6" diameter core	Ea.	B89A	15.5	1.032	1.01	37.17	7.52	45.7	71.49
For each additional inch of plank thickness in same hole, add	Ea.	B89A	960	0.017	0.13	0.61	0.12	0.86	1.28
8" diameter core	Ea.	B89A	15	1.067	1.54	38.76	7.78	48.08	74.45
For each additional inch of plank thickness in same hole, add	Ea.	B89A	768	0.021	0.2	0.75	0.16	1.11	1.64
10" diameter core	Ea.	B89A	14	1.143	1.95	41.42	8.3	51.67	80.31
For each additional inch of plank thickness in same hole, add	Ea.	B89A	640	0.025	0.25	0.9	0.19	1.34	1.98
12" diameter core	Ea.	B89A	13.5	1.185	2.4	43.01	8.61	54.02	83.81
For each additional inch of plank thickness in same hole, add	Ea.	B89A	548	0.029	0.29	1.06	0.21	1.56	2.33
Bits for core drilling, included in drilling line items									
Diamond, premium, 1" diameter	Ea.					71.69		71.69	79.05
2" diameter	Ea.					113.91		113.91	124.71
3" diameter	Ea.					153.19		153.19	168.9
4" diameter	Ea.					197.38		197.38	218
6" diameter	Ea.					304.42		304.42	333.88
8" diameter	Ea.					461.54		461.54	510.64
10" diameter	Ea.					589.2		589.2	648.12
12" diameter	Ea.					716.86		716.86	790.51
14" diameter	Ea.					834.7		834.7	918.17
18" diameter	Ea.					1153.85		1153.85	1252.05
24" diameter	Ea.					1620.3		1620.3	1792.15
Concrete Drilling									
CONCRETE IMPACT DRILLING									
Description	Unit	Crew	Daily Output	Labor Hours	Bare Material	Bare Labor	Bare Equipment	Bare Total	Total O&P
Includes bit cost, layout and set-up time, no anchors									
Up to 4" deep in concrete/brick floors/walls									
Holes, 1/4" diameter	Ea.	1 Carp	75	0.107	0.07	4.3		4.37	7.25
For each additional inch of depth in same hole, add	Ea.	1 Carp	430	0.019	0.02	0.75		0.77	1.27
3/8" diameter	Ea.	1 Carp	63	0.127	0.06	5.13		5.19	8.61
For each additional inch of depth in same hole, add	Ea.	1 Carp	340	0.024	0.01	0.95		0.96	1.6
1/2" diameter	Ea.	1 Carp	50	0.16	0.06	6.48		6.54	10.79
For each additional inch of depth in same hole, add	Ea.	1 Carp	250	0.032	0.01	1.3		1.31	2.17
5/8" diameter	Ea.	1 Carp	48	0.167	0.1	6.74		6.84	11.31
For each additional inch of depth in same hole, add	Ea.	1 Carp	240	0.033	0.03	1.35		1.38	2.27
3/4" diameter	Ea.	1 Carp	45	0.178	0.13	7.17		7.3	12.09
For each additional inch of depth in same hole, add	Ea.	1 Carp	220	0.036	0.03	1.47		1.5	2.48
7/8" diameter	Ea.	1 Carp	43	0.186	0.18	7.49		7.67	12.68
For each additional inch of depth in same hole, add	Ea.	1 Carp	210	0.038	0.05	1.54		1.59	2.61
1" diameter	Ea.	1 Carp	40	0.2	0.16	8.07		8.23	13.6
For each additional inch of depth in same hole, add	Ea.	1 Carp	190	0.042	0.04	1.7		1.74	2.86
1-1/4" diameter	Ea.	1 Carp	38	0.211	0.28	8.5		8.78	14.43
For each additional inch of depth in same hole, add	Ea.	1 Carp	180	0.044	0.07	1.79		1.86	3.06
1-1/2" diameter	Ea.	1 Carp	35	0.229	0.49	9.24		9.73	15.89
For each additional inch of depth in same hole, add	Ea.	1 Carp	165	0.048	0.12	1.95		2.07	3.39
For ceiling installations, add						40.00%			